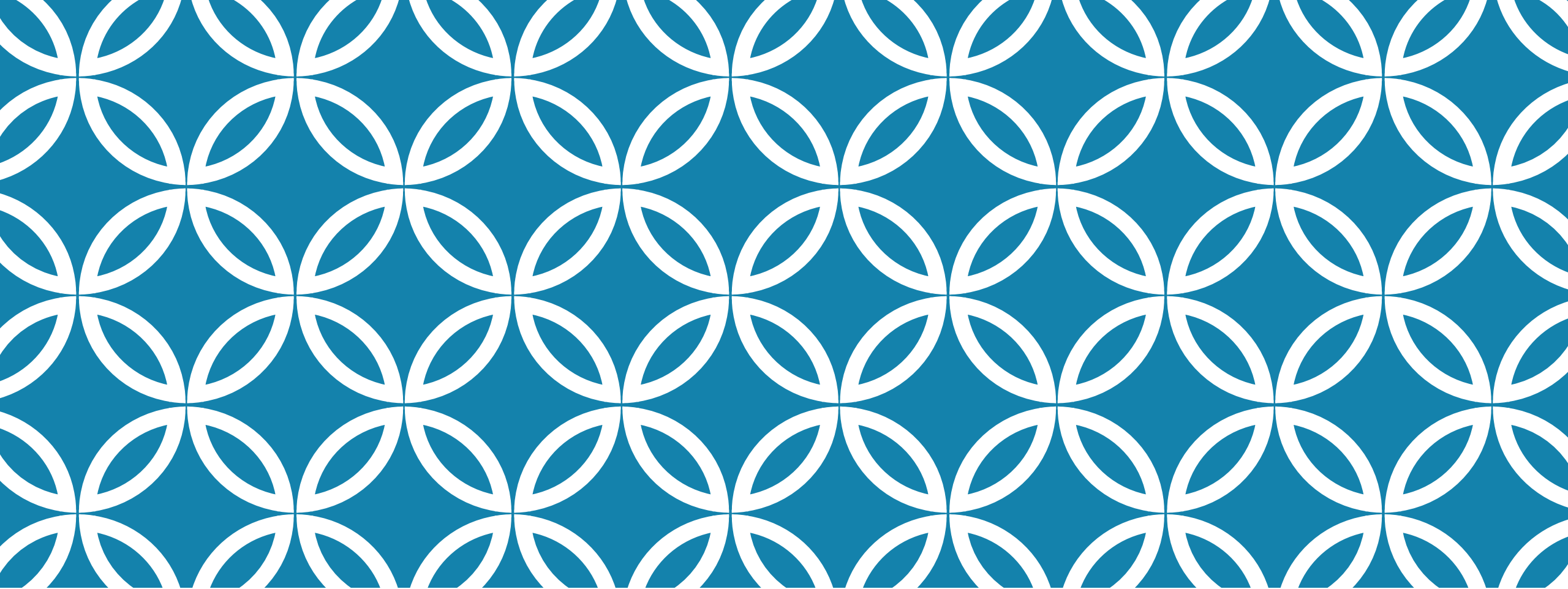




REVERSE ENGINEERING
17 CARS
IN UNDER 10 MINUTES

BRENT
STONE

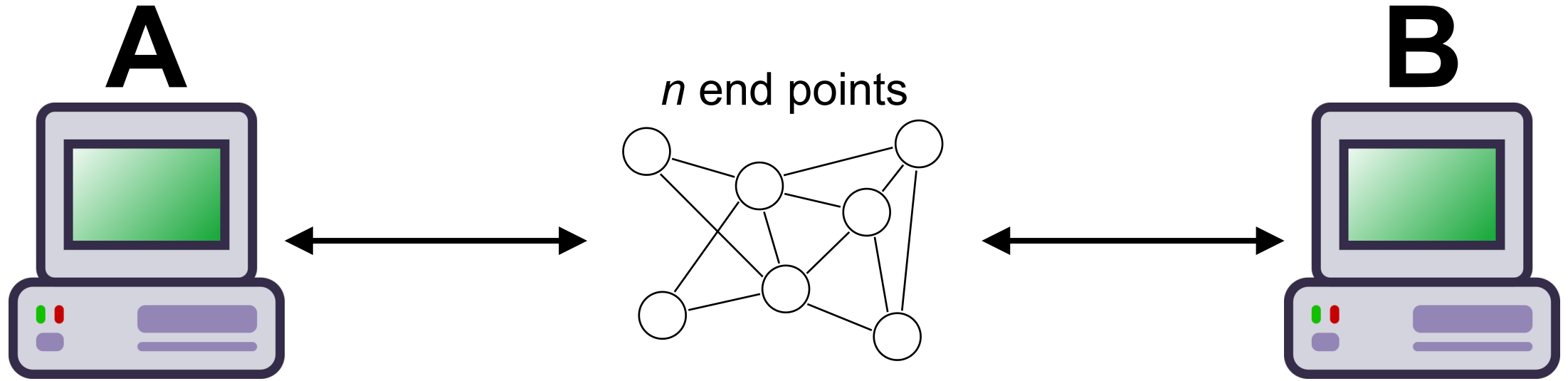
Disclaimer About This Talk and The Github Repo

The views expressed in this presentation are those of the author and do not reflect the official policy or position of the United States Air Force, the United States Army, the United States Department of Defense or the United States Government. The material publicly released on https://github.com/brent-stone/CAN_Reverse_Engineering/, up to and including commit ac0e55f on 26 March 2019, is declared a work of the U.S. Government and is not subject to copyright protection in the United States.

APPROVED FOR PUBLIC RELEASE; DISTRIBUTION UNLIMITED

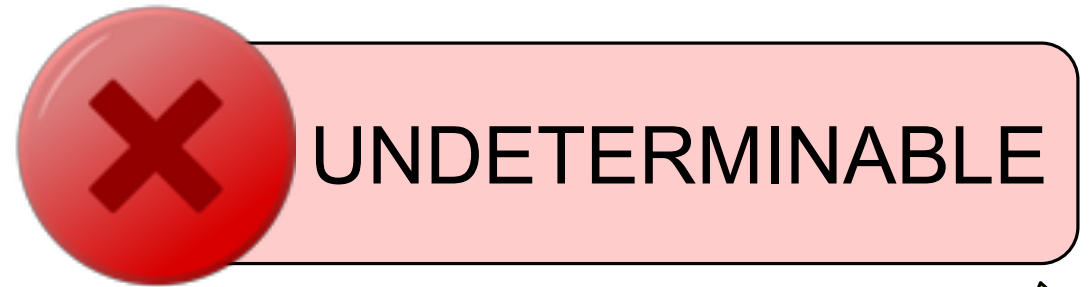
Case Numbers: 88ABW-2019-0910, 88ABW-2019-0024

General Use Networks



- Modify End Points
- Modify Routing

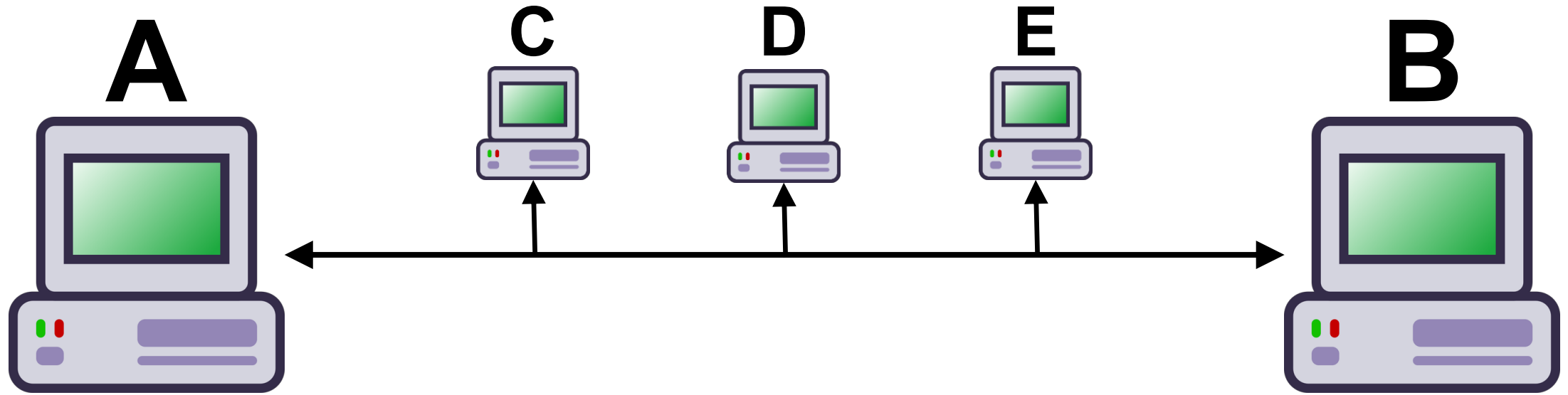
Meta
Data



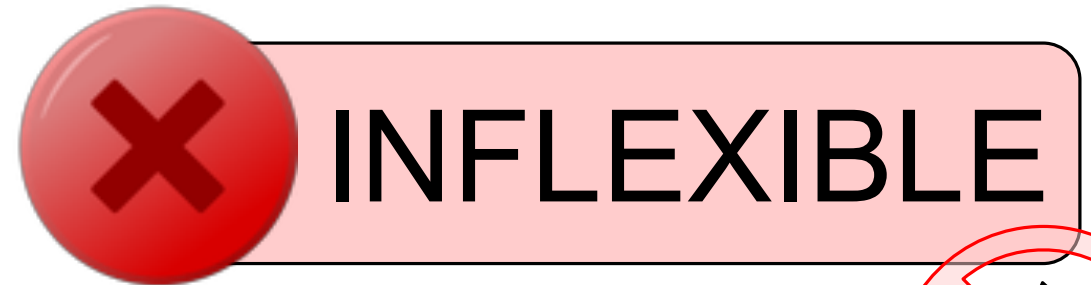
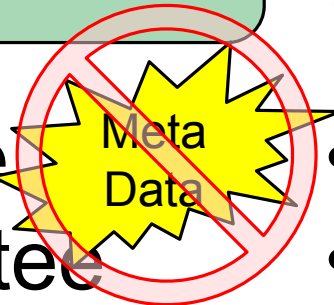
- No delivery guarantee
- No timeliness guarantee

Meta
Data

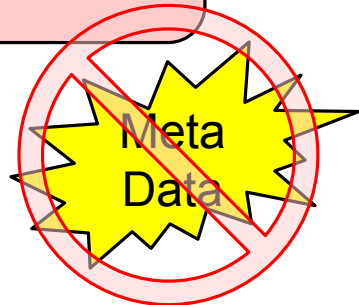
Control Networks



- Delivery Guarantee
- Timeliness Guarantee



- Fixed End Points
- Fixed Routing



interwebs

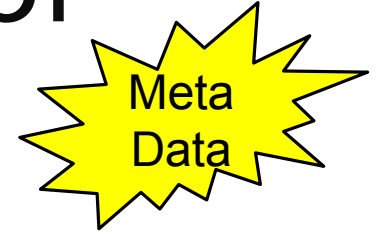
VoIP

**Lots of people helping
others play with
general use networks...**

IoT

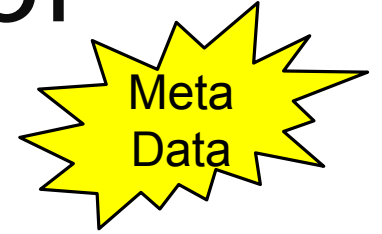
smart phones

Automated Reverse Engineering of General Use Networks



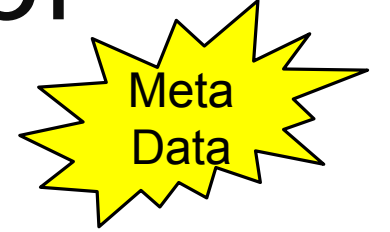
1. P. Ducange, G. Mannara, F. Marcelloni, R. Pecori, and M. Vecchio, "A novel approach for internet traffic classification based on multi-objective evolutionary fuzzy classifiers," in 2017 IEEE International Conference on Fuzzy Systems (FUZZ-IEEE), 2017, pp. 1-6.
2. J. Yuan, Z. Li, and R. Yuan, "Information entropy based clustering method for unsupervised internet traffic classification," in IEEE International Conference on Communications (ICC), 2008, pp. 1588-1592.
3. C. Besiktas and H. A. Mantar, "Real-Time Traffic Classification Based on Cosine Similarity Using Sub-application Vectors," in Proceedings of the Traffic Monitoring and Analysis 4th International Workshop, 2012, vol. 7189, pp. 89-92.
4. A. Trifilo, S. Burschka, and E. Biersack, "Traffic to protocol reverse engineering," in IEEE Symposium on Computational Intelligence for Security and Defense Applications (CISDA), 2009, pp. 1-8.
5. M. E. DeYoung, "Dynamic protocol reverse engineering: a grammatical inference approach," Air Force Institute of Technology, 2008.
6. W. Cui, M. Peinado, K. Chen, H. J. Wang, and L. Irwin-Briz, "Tupni: Automatic Reverse Engineering of Input Formats," in 15th ACM Conference on Computer and Communications Security (CCS), 2008, pp. 391-402.
7. J. Newsome, D. Brumley, J. Franklin, and D. Song, "Replayer: automatic protocol replay by binary analysis," in 13th ACM conference on Computer and Communications Security (CCS), 2006, p. 311.
8. J. Caballero, P. Poosankam, C. Kreibich, and S. D., "Dispatcher: Enabling active botnet infiltration using automatic protocol reverse-engineering," in 16th ACM Conference on Computer and Communications Security (CCS), 2009, pp. 621-634.
9. J. Caballero, H. Yin, Z. Liang, and D. Song, "Polyglot: Automatic Extraction of Protocol Message Format using Dynamic Binary Analysis," in 14th ACM Conference on Computer and Communications Security (CCS), 2007, pp. 317-329.
10. W. Cui, V. Paxson, N. C. Weaver, and R. H. Katz, "Protocol-Independent Adaptive Replay of Application Dialog," in Network and Distributed System Security Symposium (NDSS), 2006, pp. 279-293.

Automated Reverse Engineering of General Use Networks



11. M. Wakchaure, S. Sarwade, I. Siddavatam, and P. Range, "Reconnaissance of Industrial Control System By Deep Packet Inspection," in 2nd IEEE International Conference on Engineering and Technology (ICETECH), 2016, no. 3, pp. 1093-1096.
12. J. Antunes, N. Neves, and P. Verissimo, "Reverse engineering of protocols from network traces," in 18th Working Conference on Reverse Engineering, 2011, pp. 169-178.
13. M. A Beddoe, "Network protocol analysis using bioinformatics algorithms," McAfee, Santa Clara, CA, USA, 1, 2004.
14. Y. Wang, Z. Zhang, D. Yao, B. Qu, and L. Guo, "Inferring Protocol State Machine from Network Traces: A Probabilistic Approach," in International Conference on Applied Cryptography and Network Security, 2011, pp. 1-18.
15. P. M. Comparetti, G. Wondracek, C. Kruegel, and E. Kirda, "Prospex: Protocol specification extraction," in IEEE Symposium on Security and Privacy, 2009, pp. 110-125.
16. J. Erman and M. Arlitt, "Traffic classification using clustering algorithms," in 2006 SIGCOMM Workshop on Mining Network Data, 2006, pp. 281-286.
17. F. Alam, R. Mehmood, I. Katib, and A. Albeshri, "Analysis of Eight Data Mining Algorithms for Smarter Internet of Things (IoT)," in International Workshop on Data Mining in IoT Systems (DaMIS 2016), 2016, vol. 98, no. 1, pp. 437-442.
18. Y. Wang et al., "A semantics aware approach to automated reverse engineering unknown protocols," in 20th IEEE International Conference on Network Protocols (ICNP), 2012, pp. 1-10.
19. J. Roning, "PROTOS Protocol Genome Project," Oulu University Secure Programming Group, 2010. [Online]. Available: <https://www.ee.oulu.fi/roles/ouspg/genome>. [Accessed: 01-Jan-2017].
20. R. L. S. Puupera, "Domain Model Based Black Box Fuzzing Using Regular Languages," University of Oulu, 2010.
21. K. Choi, Y. Son, J. Noh, H. Shin, J. Choi, and Y. Kim, "Dissecting Customized Protocols: Automatic Analysis for Customized Protocols Based on IEEE 802.15.4," in 9th International Conference on Security of Information and Networks, 2016, pp. 183-193.

Automated Reverse Engineering of General Use Networks

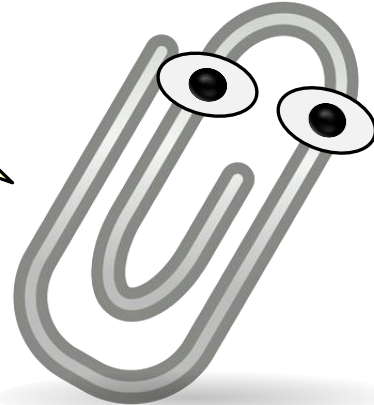


- 22. Y. Wang, Y. Xiang, J. Zhang, and S. Yu, "A novel semi-supervised approach for network traffic clustering," in 5th International Conference on Network and System Security (NSS), 2011, pp. 169-175.
- 23. W. Cui, J. Kannan, and H. J. Wang, "Discoverer: Automatic Protocol Reverse Engineering from Network Traces," in USENIX Security, 2007, no. 2, pp. 199-212.
- 24. J. Zhang, C. Chen, Y. Xiang, and W. Zhou, "Semi-supervised and compound classification of network traffic," in Proceedings 32nd IEEE International Conference on Distributed Computing Systems Workshops (ICDCSW), 2012, pp. 617-621.
- 25. T. Glennan, C. Leckie, and S. M. Erfani, "Improved Classification of Known and Unknown Network Traffic Flows Using Semi-supervised Machine Learning," in 21st Australasian Conference on Information Security and Privacy (ACISP), 2016, vol. 2, pp. 493-501.

But what about robots, cars, and other control networks?

Now your computer can help!

Hi! Do you need
assistance?



#Started canhandler on can0

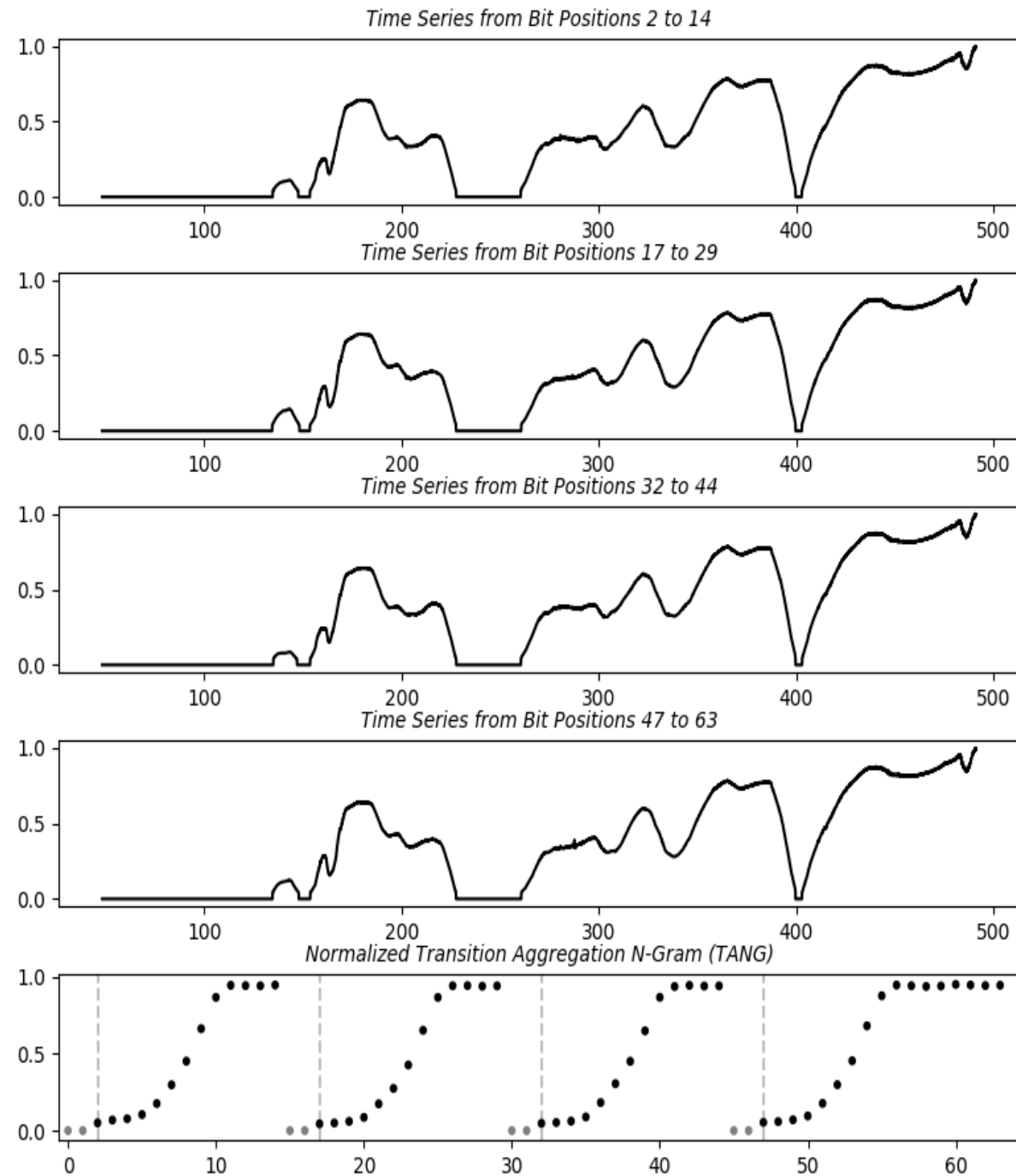
#Setup complete: 48.7387

#Format:

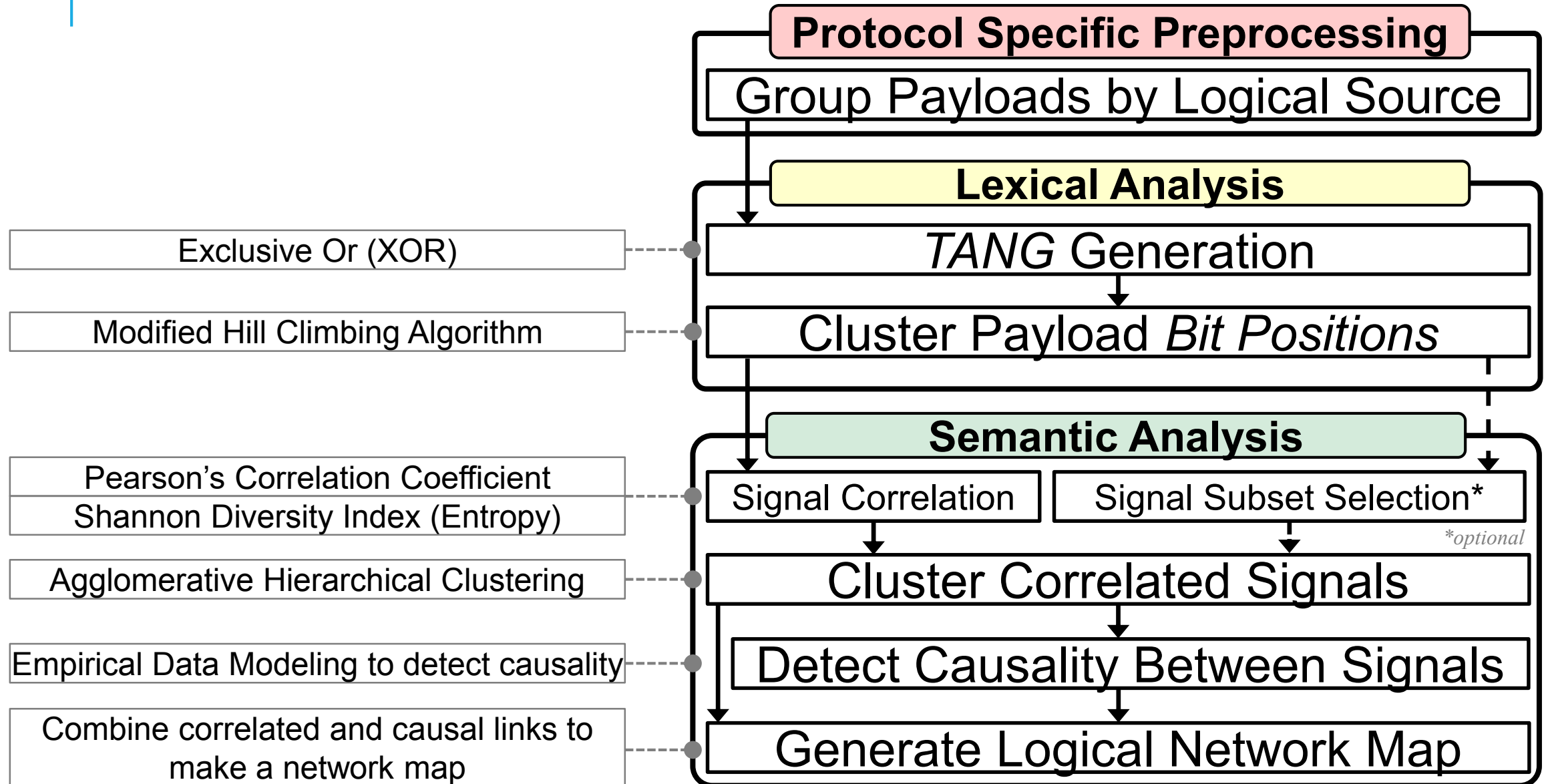
Time:	ID	DLC	Data
48.740:	4a8 8	00	00 00 40 00 00 00 00
48.740:	020 7	00	00 07 01 00 00 2f
48.742:	0b4 8	00	00 00 00 ac 00 00 68
48.742:	025 8	00	11 00 00 78 78 78 a6
48.743:	024 8	02	00 00 08 62 04 81 1f
48.743:	235 6	00	00 00 00 3d
48.744:	499 8	00	00 00 00 00 00
48.745:	49a 8	00	85 80 28 a8
48.746:	49b 8	00	a0 00 48 10
48.746:	262 5	20	00 00 89
48.747:	49d 8	61	60 00 9d 19 c6 c5
48.747:	1c4 8	00	00 00 00 00 00 cd
48.749:	0aa 8	1a	6f 1a 6f 1a 6f 6f
48.749:	0b6 4	00	00 00 00 ba
48.749:	224 8	00	00 00 00 00 00 08
48.751:	127 8	68	10 00 08 00 0c ed a9
48.751:	020 7	00	00 07 01 00 00 2f
48.751:	230 7	d4	43 00 00 00 50
48.752:	025 8	00	11 00 00 82 82 82 c4

.....

Time Series and TANG for Arbitration ID 0x1d0

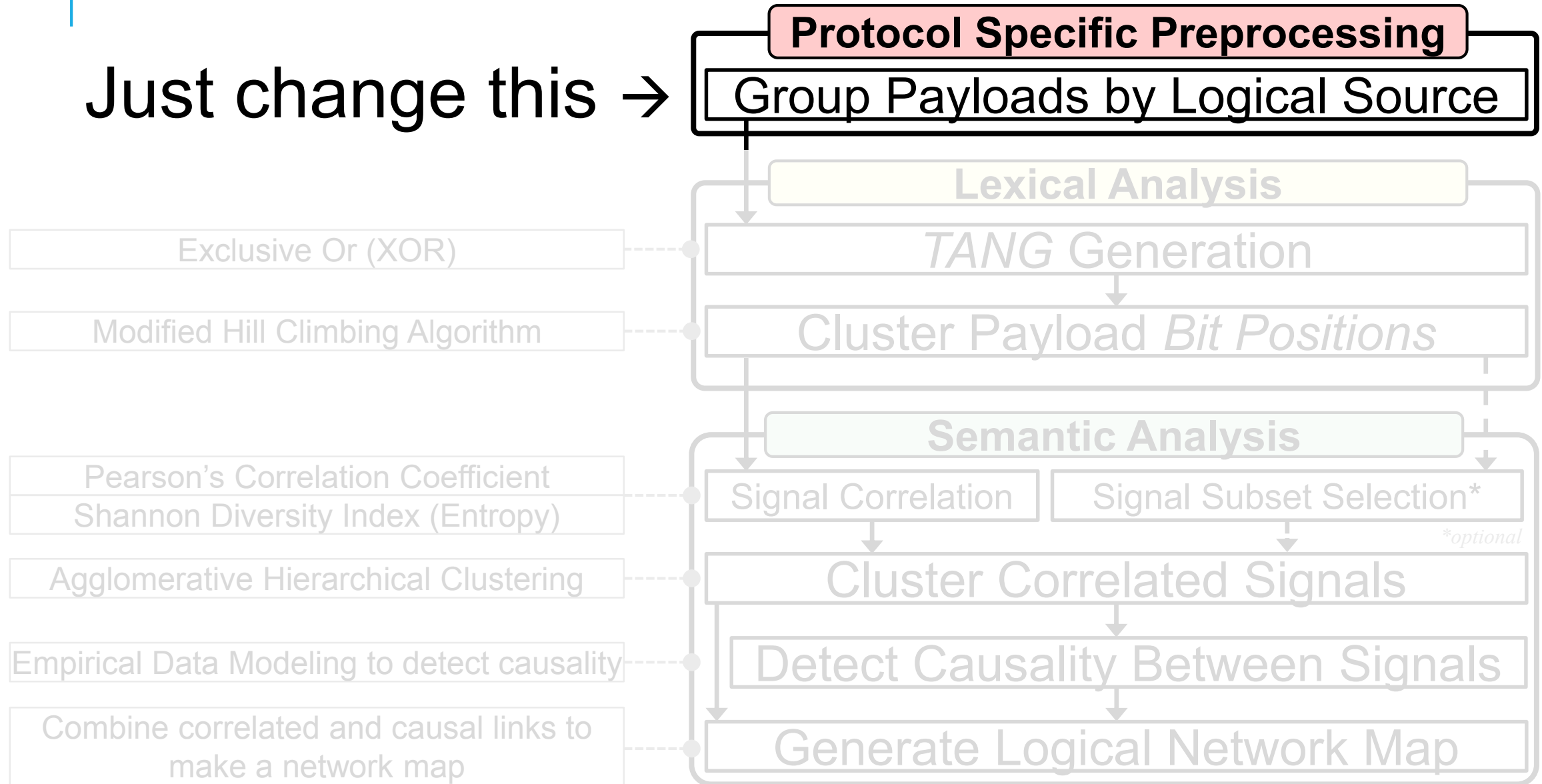


Code on GitHub does this...

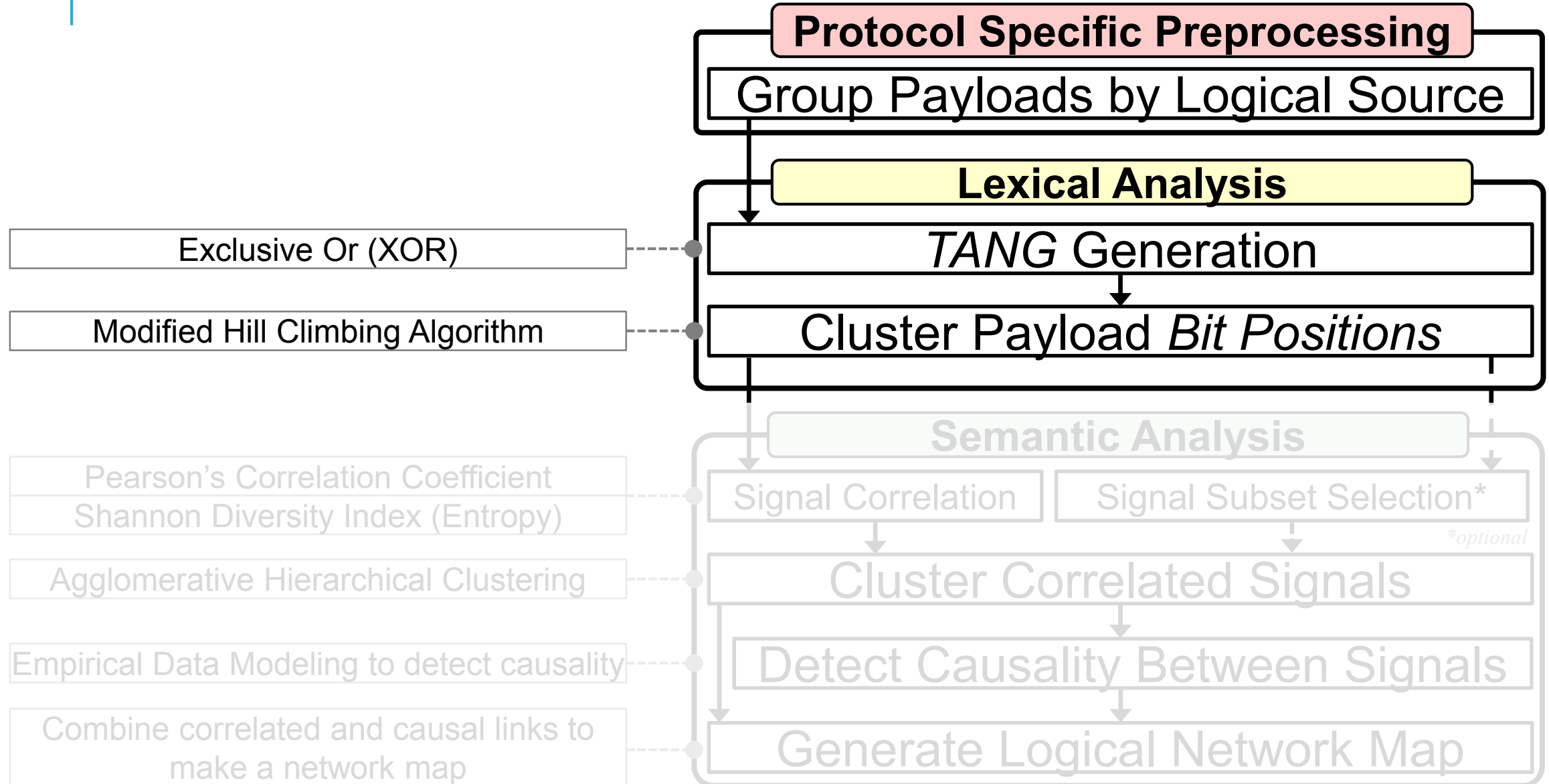


Different Control Network Protocol?

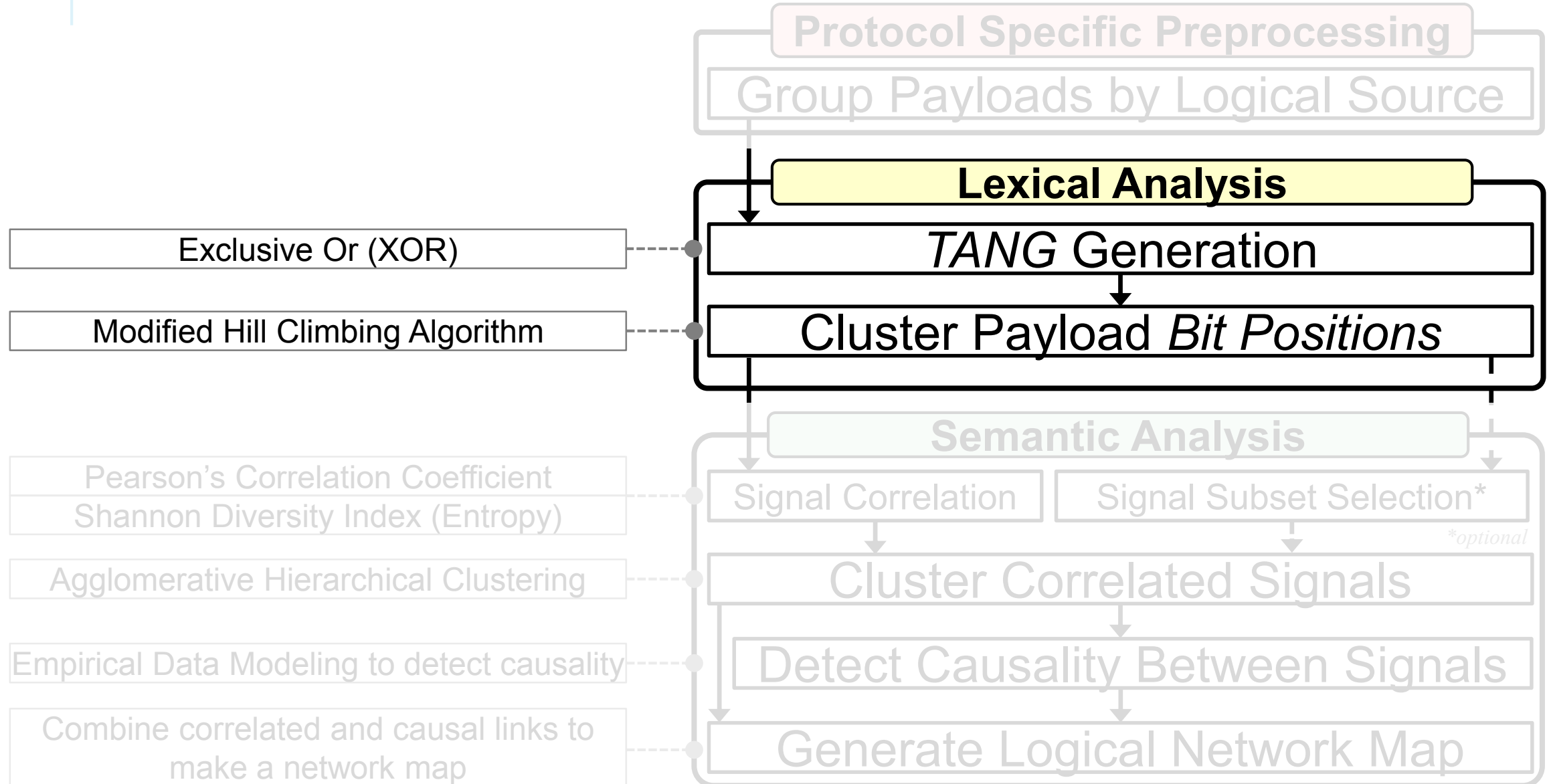
Just change this →



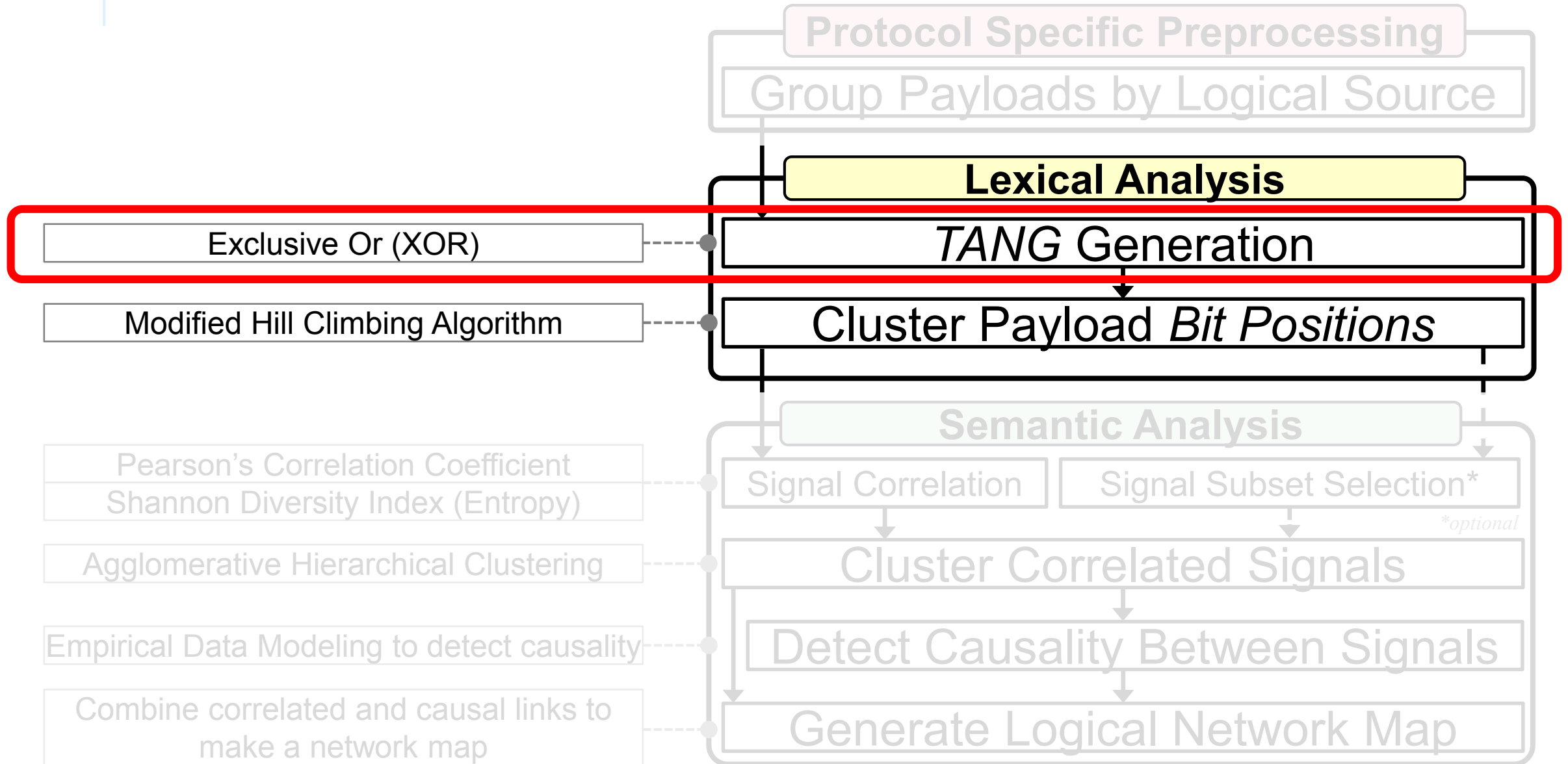
The demo is doing this...



I'll walk you through this...



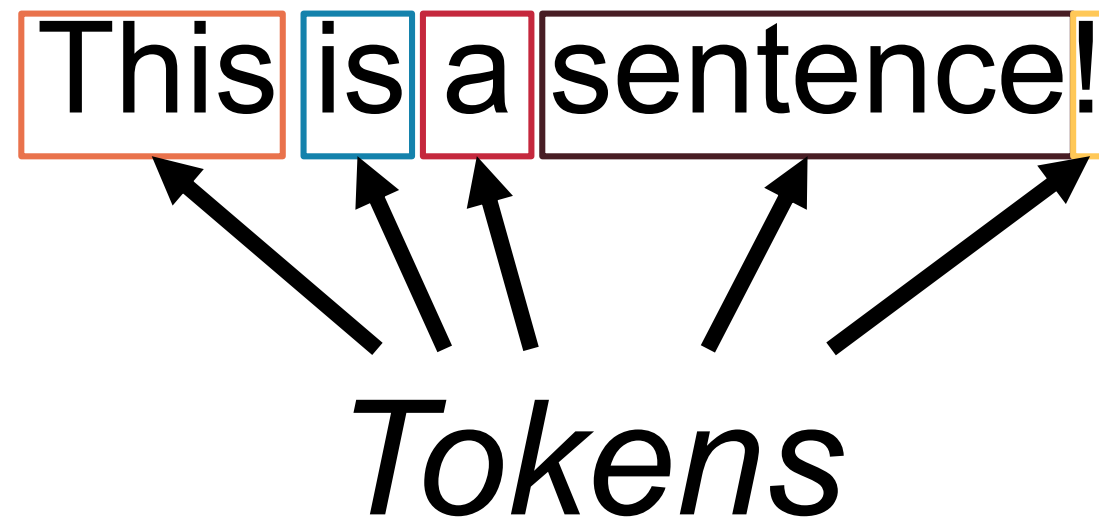
Unsupervised Reverse Engineering



Lexical & Semantic Analysis

This is a sentence!

Lexical Analysis



Semantic Analysis

This is a sentence!
noun

Token ↑
Type

Lexical Analysis

Payload Tokenization

64-bit Payloads

Time	Bit 0	Bit 63
48.45	1	0
48.95	1	0
49.46	1	0
49.96	0	0
50.46	0	0
50.96	1	0
⋮	⋮	⋮

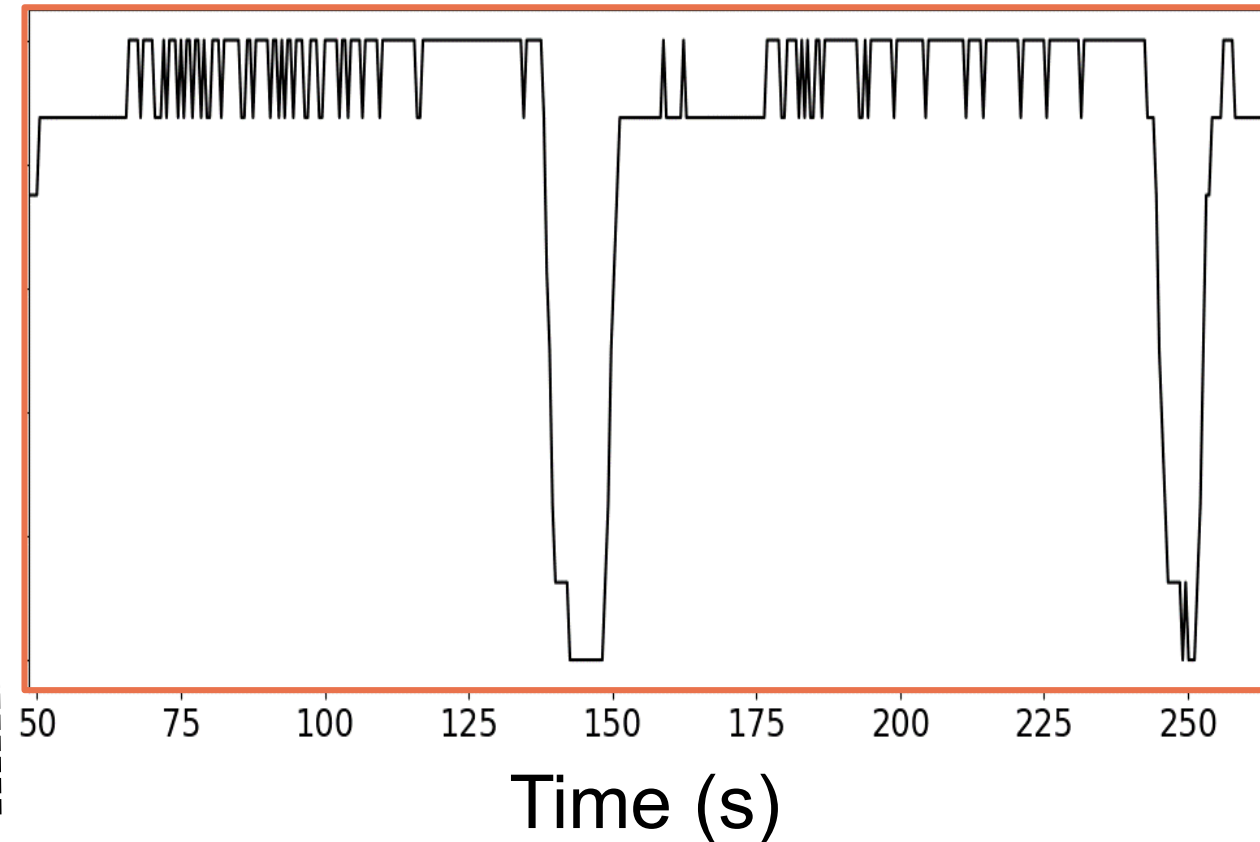
This is a sentence!

Lexical Analysis

Payload Tokenization

64-bit Payloads

Time	Bit 0	Bit 63
48.45	1	0
48.95	1	0
49.46	1	0
49.96	0	0
50.46	0	0
50.96	1	0
⋮	⋮	⋮



Lexical Analysis

Payload Tokenization

64-bit Payloads

Time	Bit 0	Bit 63
48.45	1	0
48.95	1	0
49.46	1	0
49.96	0	0
50.46	0	0
50.96	1	0
⋮	⋮	⋮

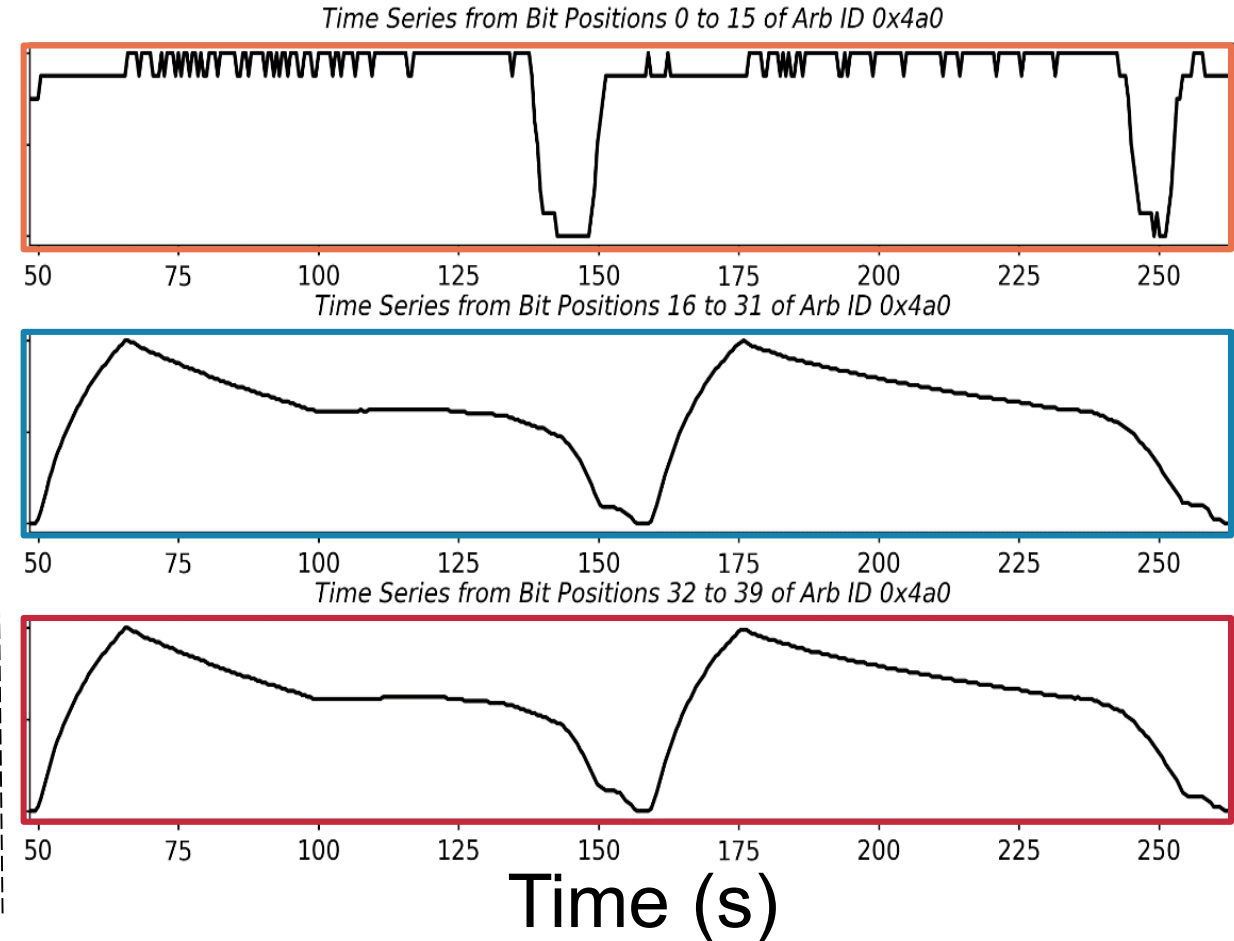
This is a sentence!

Lexical Analysis

Payload Tokenization

64-bit Payloads

Time	Bit 0	Bit 63
48.45	1	0
48.95	1	0
49.46	1	0
49.96	0	0
50.46	0	0
50.96	1	0
⋮	⋮	⋮



Payload Tokenization

By Least Significant Bit

Bit Position: 0 1 2 3 4 5 6 7 8 9

**Observed
Payloads**

0	1	1	1	0	0	0	0	0	0
1	0	0	0	0	0	0	0	0	1
1	0	0	1	0	0	0	0	1	0
1	0	1	0	0	0	0	0	1	1
1	0	1	1	0	0	0	1	0	0
1	1	0	0	0	0	0	1	0	1
1	1	0	1	0	0	0	1	1	0
1	1	1	0	0	0	0	1	1	1

Payload Tokenization

By Least Significant Bit

Bit Position: 0 1 2 3 4 5 6 7 8 9

Exclusive-OR gate



A	B	Output
0	0	0
0	1	1
1	0	1
1	1	0

0	1	1	1	0	0	0	0	0	0
1	0	0	0	0	0	0	0	0	0
1	0	0	1	0	0	0	0	1	0
1	0	1	0	0	0	0	0	0	0
1	0	0	0	0	0	0	1	0	0
1	1	0	0	0	0	0	1	0	0
1	1	0	0	0	0	0	1	1	0

Payload Tokenization

By Least Significant Bit

Bit Position: 0 1 2 3 4 5 6 7 8 9

Exclusive-OR gate



A	B	Output
0	0	0
0	1	1
1	0	1
1	1	0

1	1	1	1	0	0	0	0	0	1
0	0	0	1	0	0	0	0	1	1
0	0	1	1	0	0	0	0	0	1
0	0	0	1	0	0	0	1	1	1
0	1	1	1	0	0	0	0	0	1
0	0	0	1	0	0	0	0	1	1
0	0	1	1	0	0	0	0	0	1

Payload Tokenization

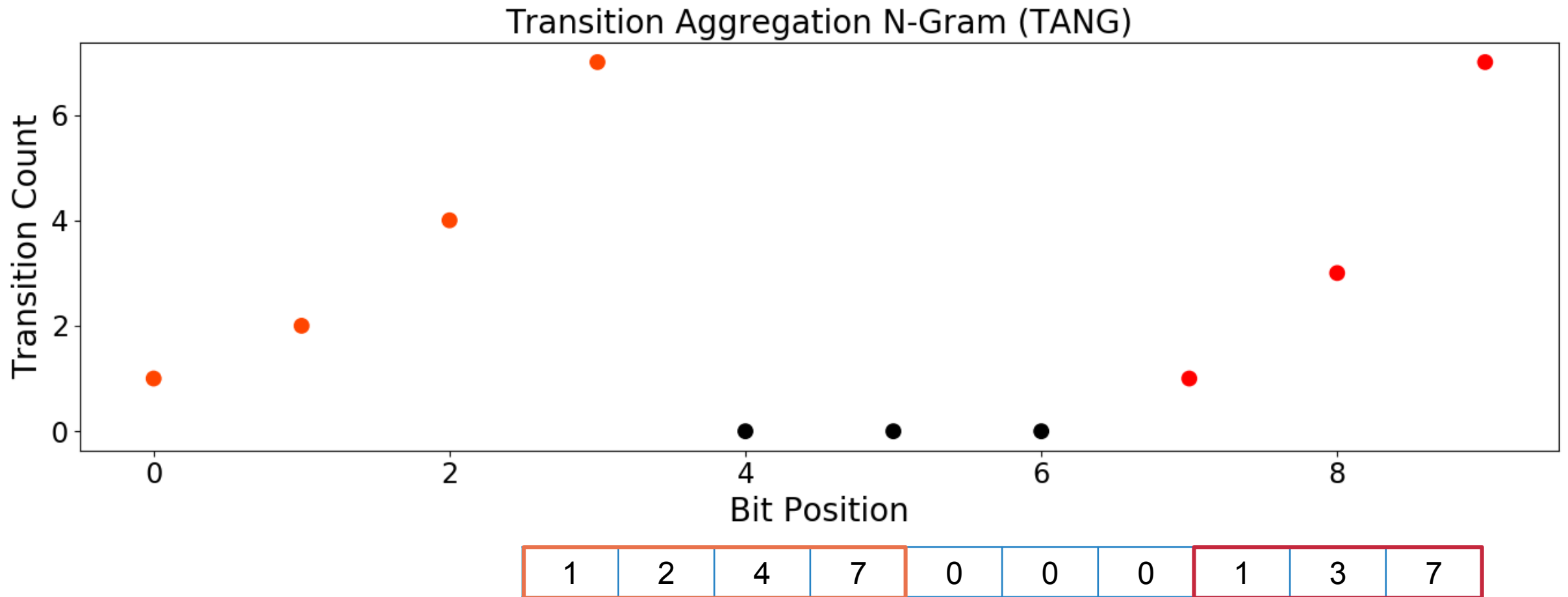
By Least Significant Bit

Bit Position: 0 1 2 3 4 5 6 7 8 9

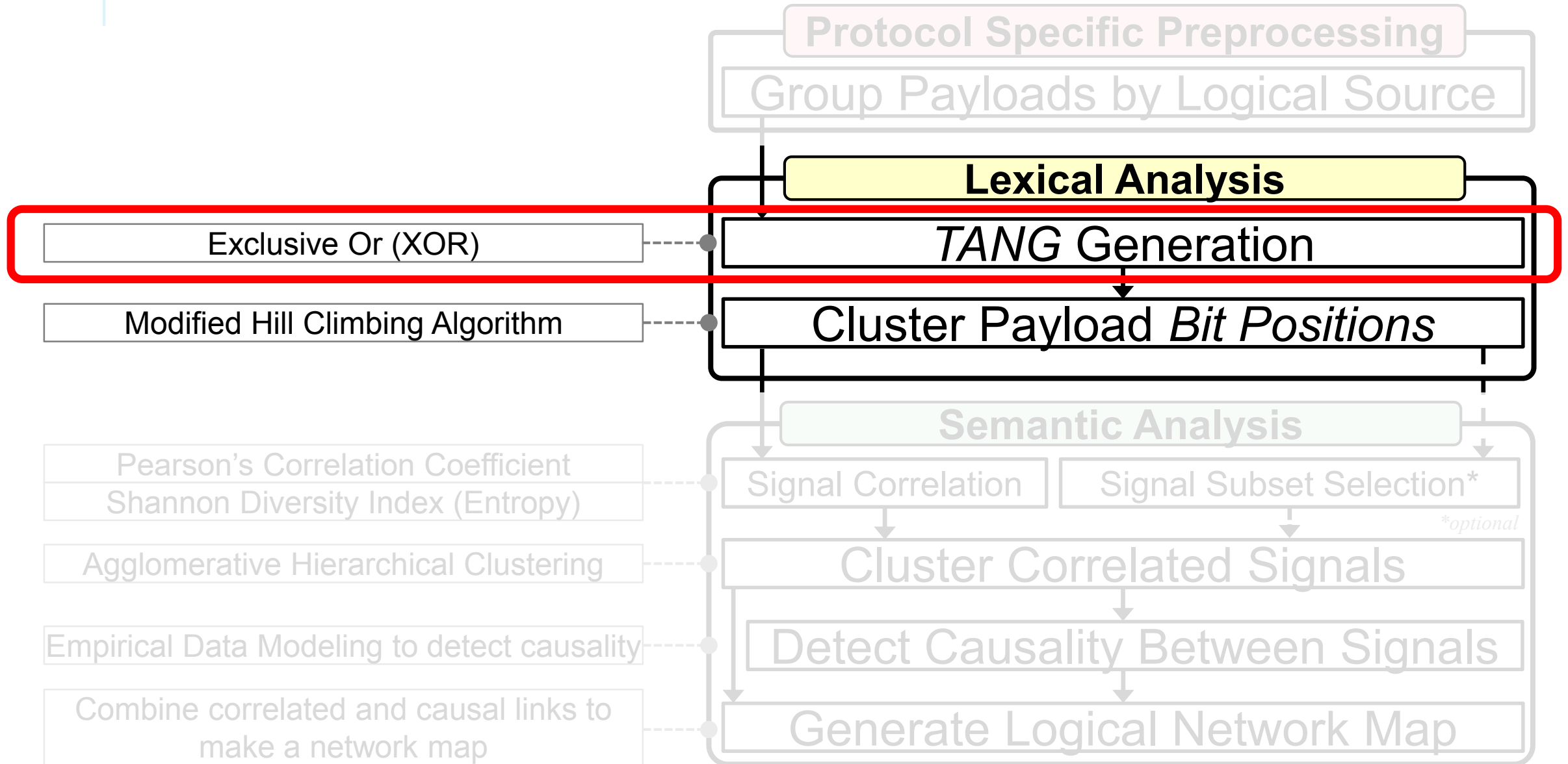
	1	1	1	1	0	0	0	0	0	1
	0	0	0	1	0	0	0	0	1	1
	0	0	1	1	0	0	0	0	0	1
	0	0	0	1	0	0	0	1	1	1
	0	1	1	1	0	0	0	0	0	1
	0	0	0	1	0	0	0	0	1	1
+	0	0	1	1	0	0	0	0	0	1
	1	2	4	7	0	0	0	1	3	7

Payload Tokenization

By Least Significant Bit

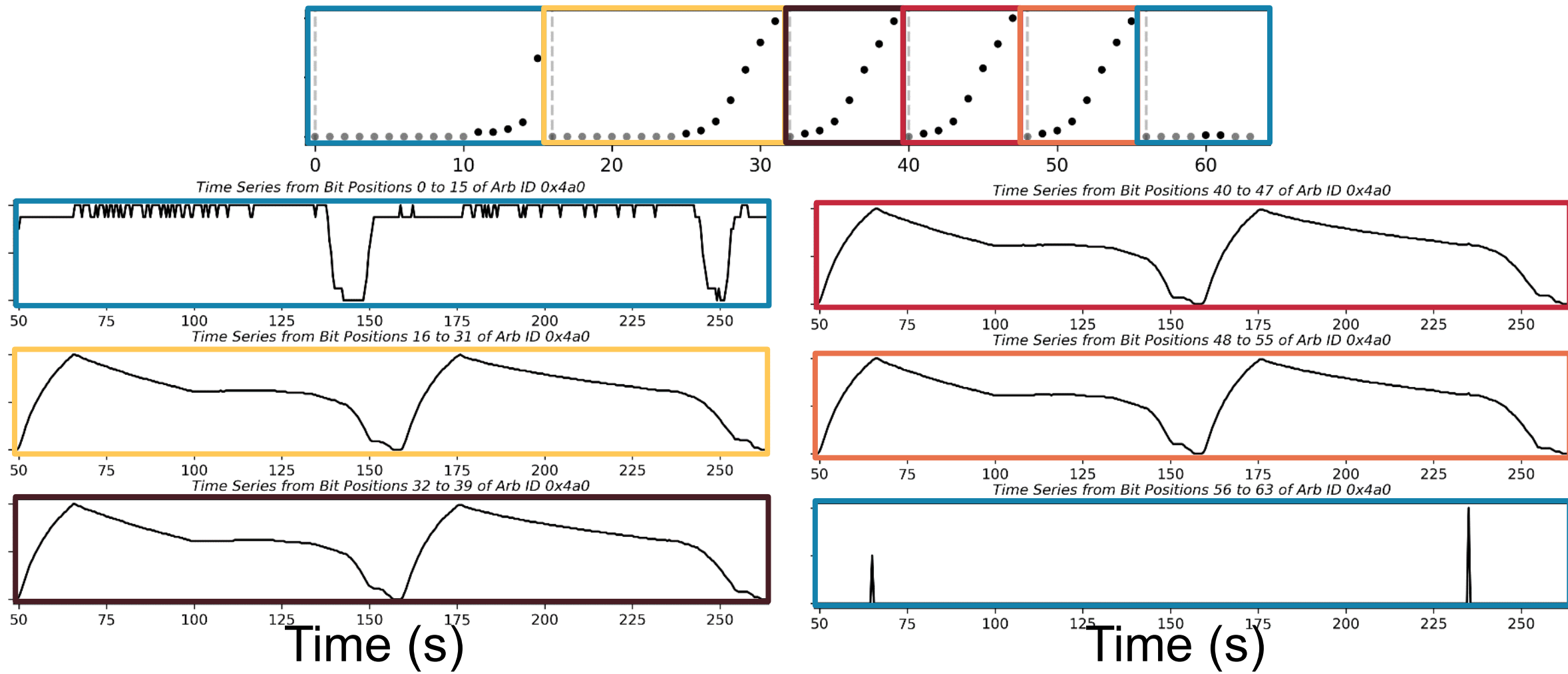


Unsupervised Reverse Engineering

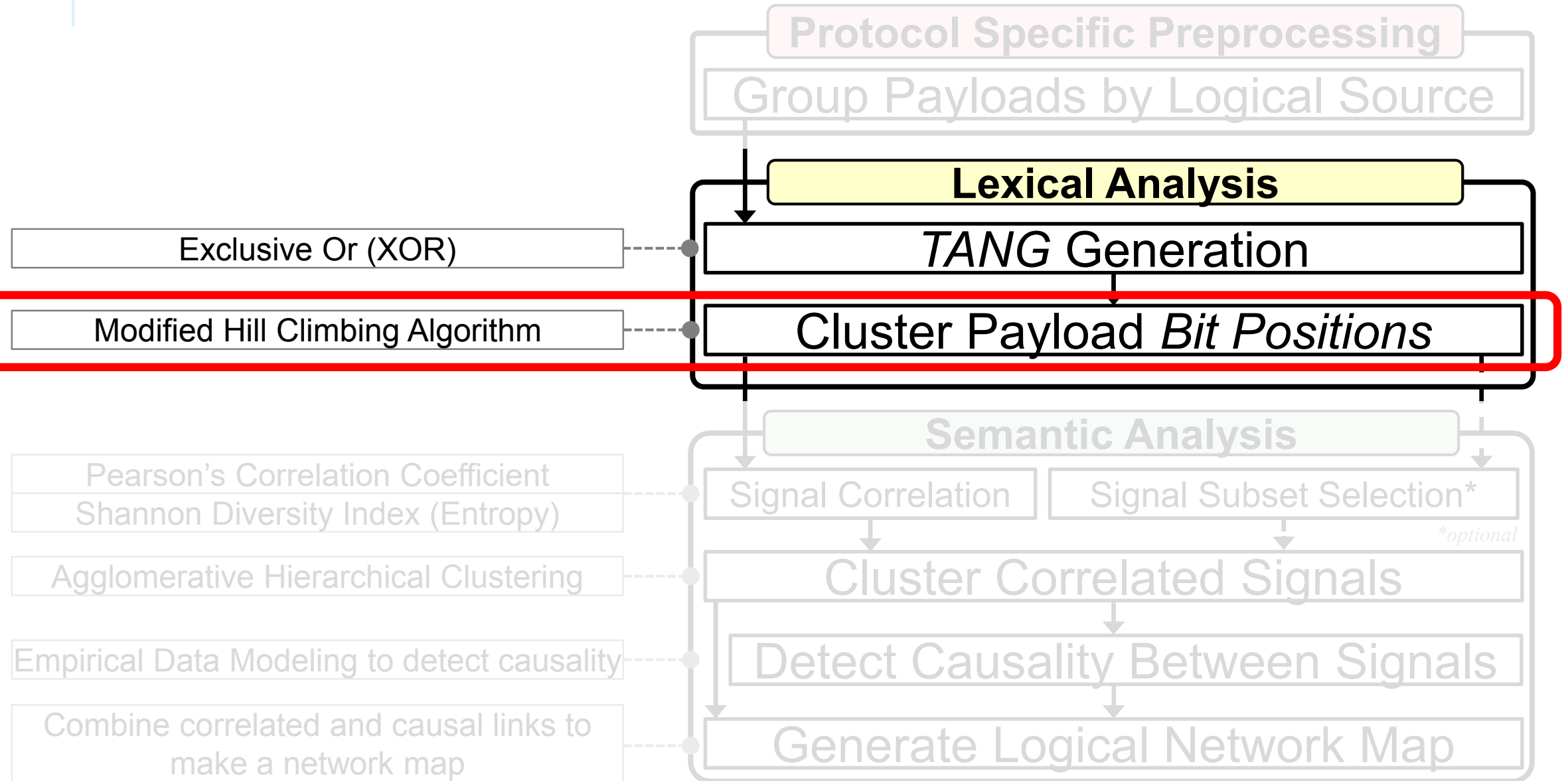


Payload Tokenization

By Least Significant Bit

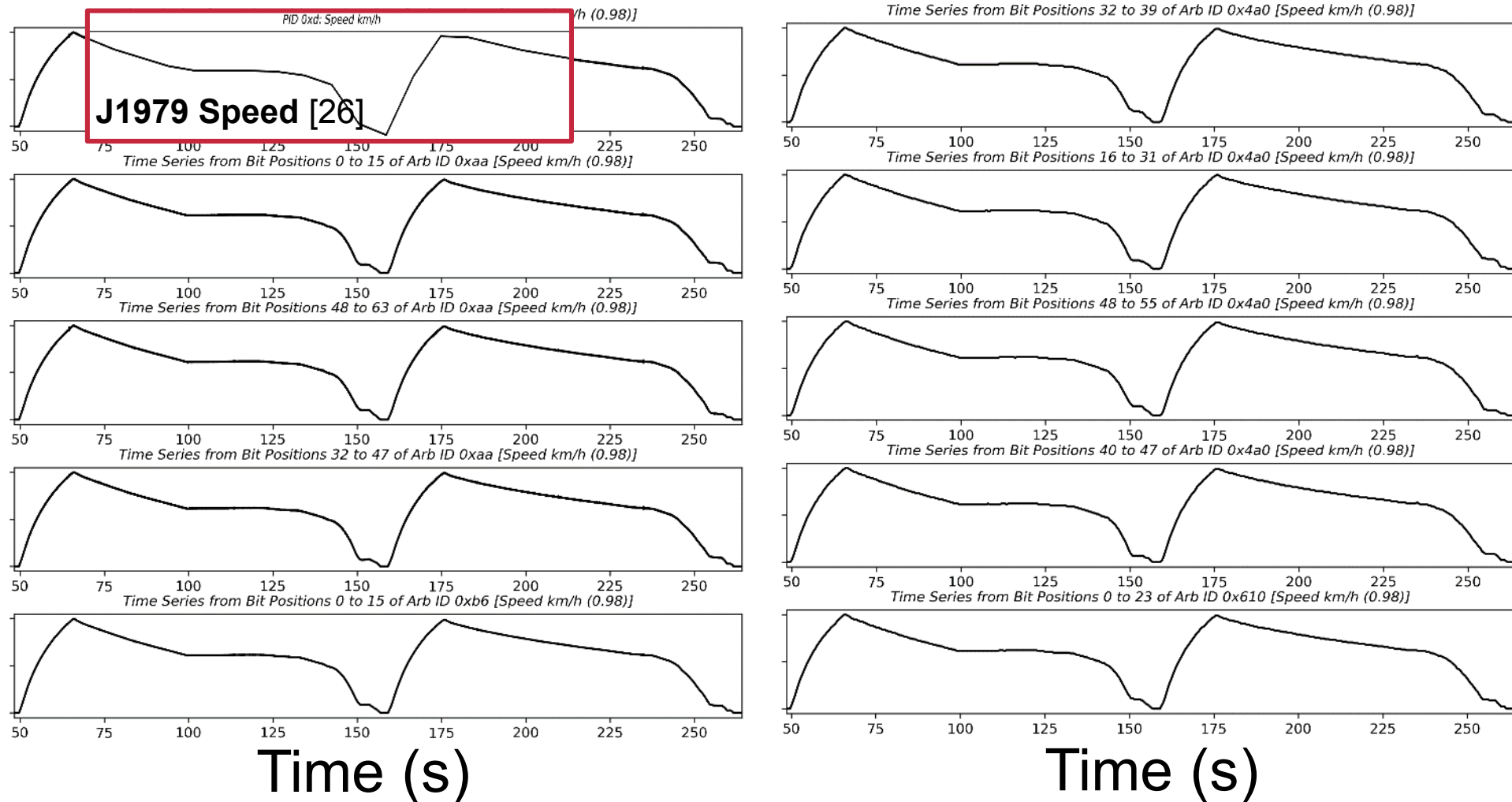


Unsupervised Reverse Engineering



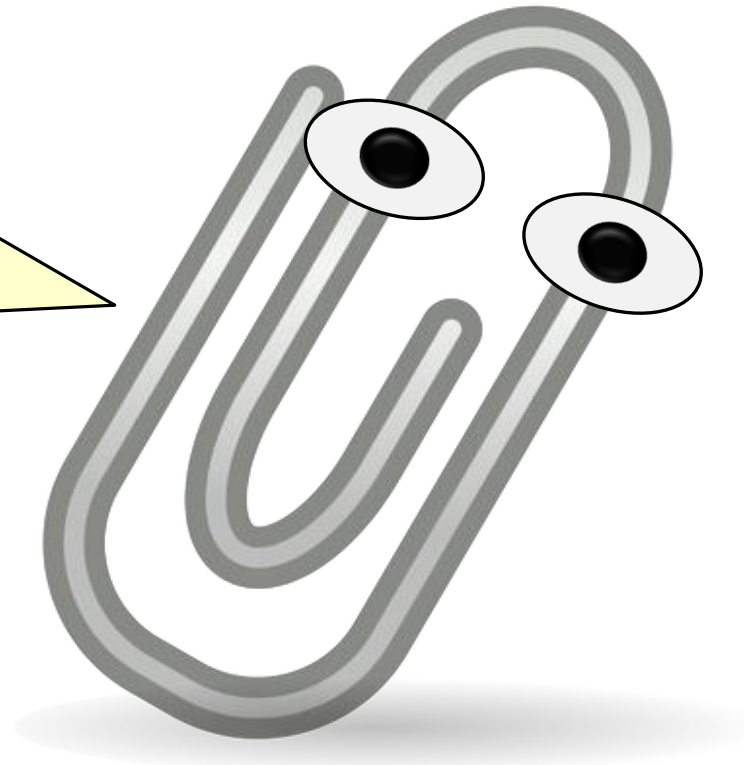
Semantic Analysis

Correlated and Causal Relationships



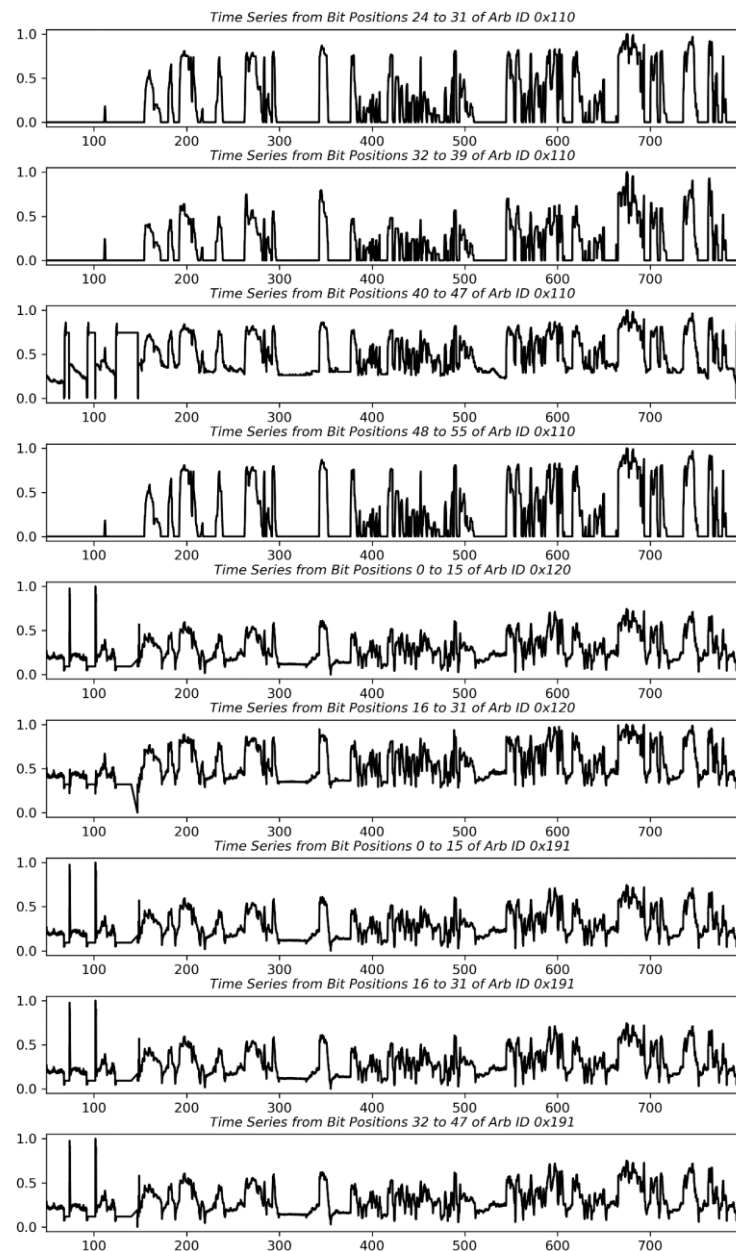
SHOW ME WHAT YOU GOT!

Let's reverse
engineer
some cars!

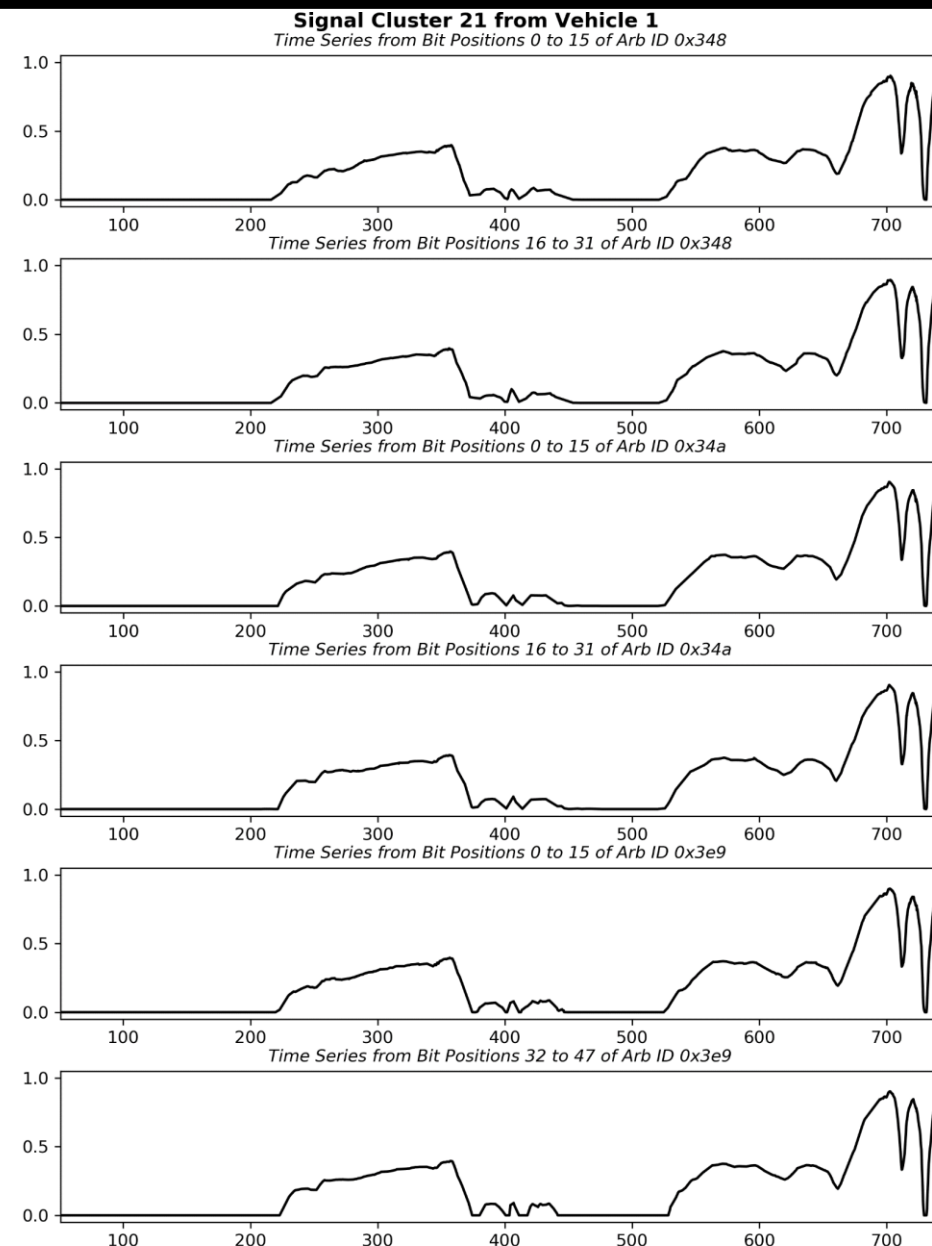


https://github.com/brent-stone/CAN_Reverse_Engineering

VEHICLE 1



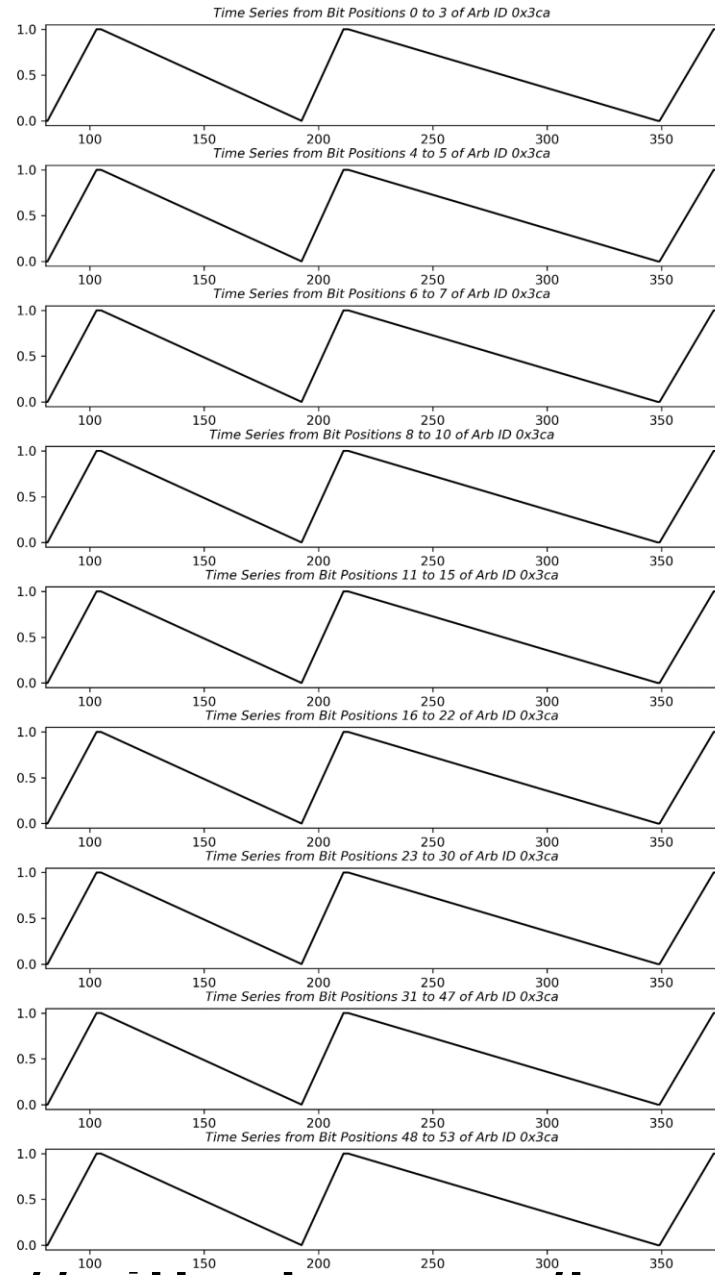
VEHICLE 2



**CROPPED
TO FIT ON
SLIDE**

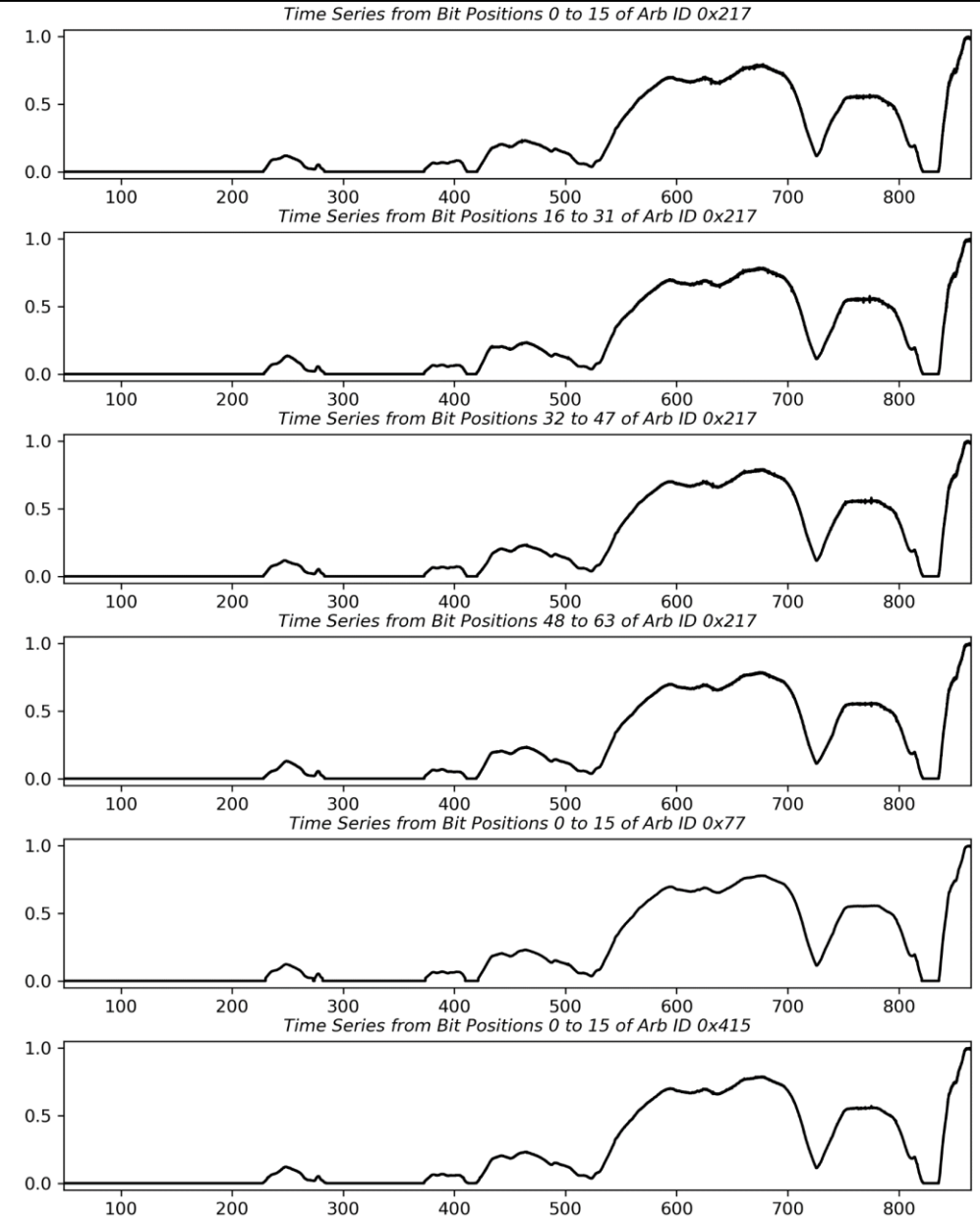
https://github.com/brent-stone/CAN_Reverse_Engineering

VEHICLE 3



**CROPPED
TO FIT ON
SLIDE**

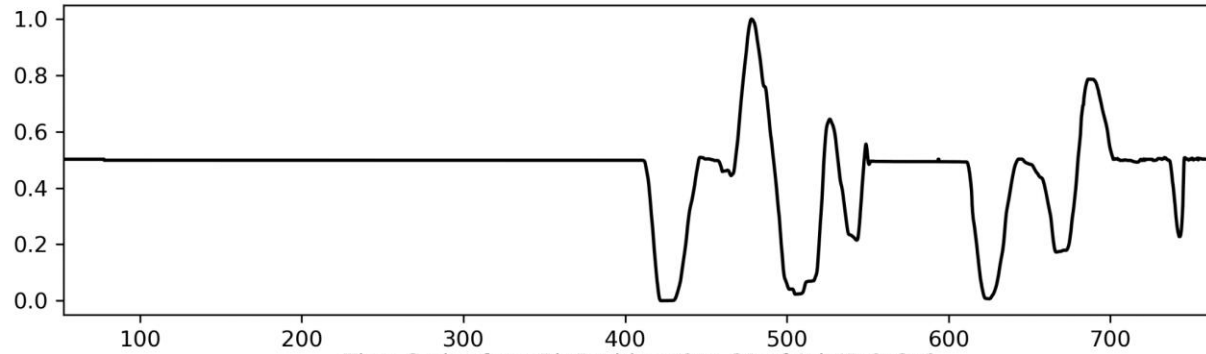
VEHICLE 4



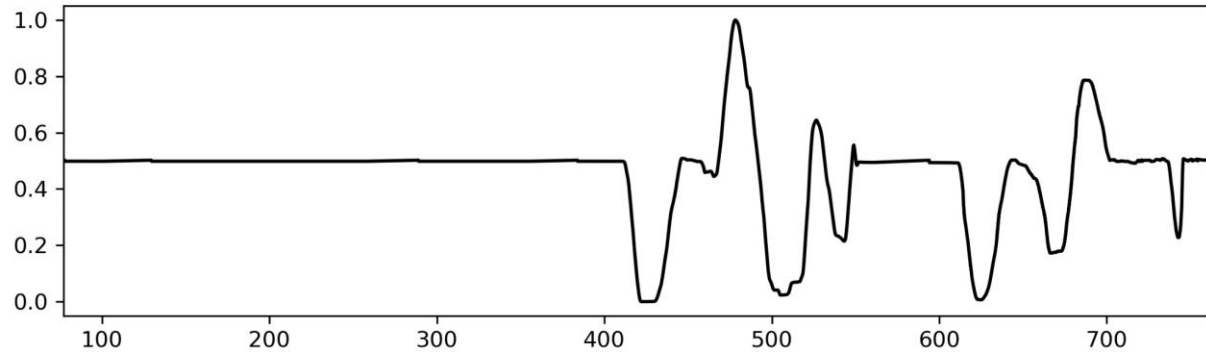
https://github.com/brent-stone/CAN_Reverse_Engineering

VEHICLE 5

Time Series from Bit Positions 0 to 15 of Arb ID 0x76

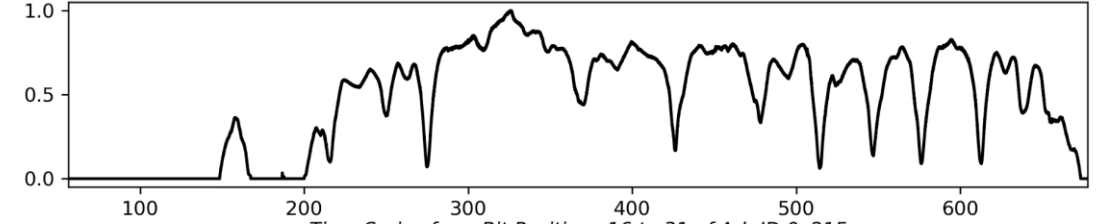


Time Series from Bit Positions 0 to 31 of Arb ID 0x3a8

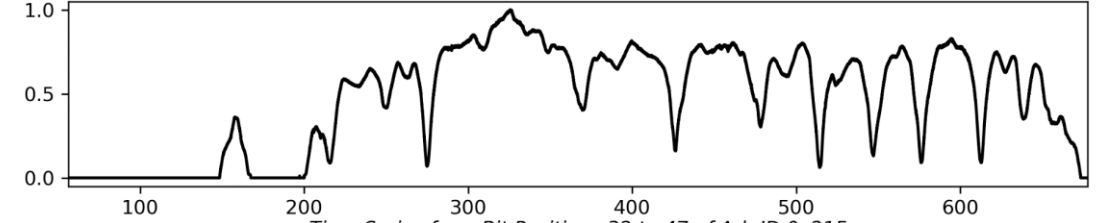


VEHICLE 6

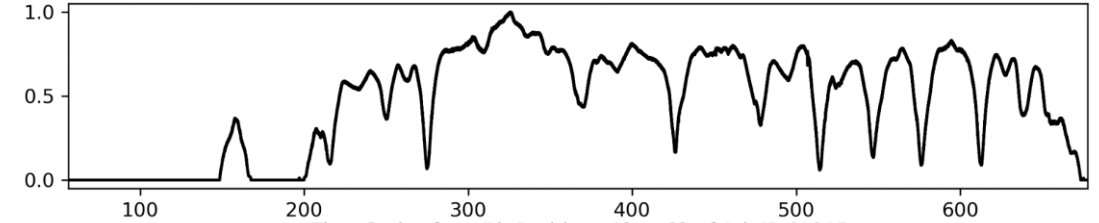
Time Series from Bit Positions 0 to 15 of Arb ID 0x215



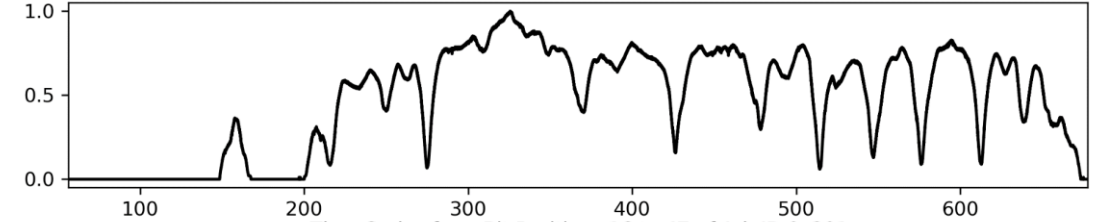
Time Series from Bit Positions 16 to 31 of Arb ID 0x215



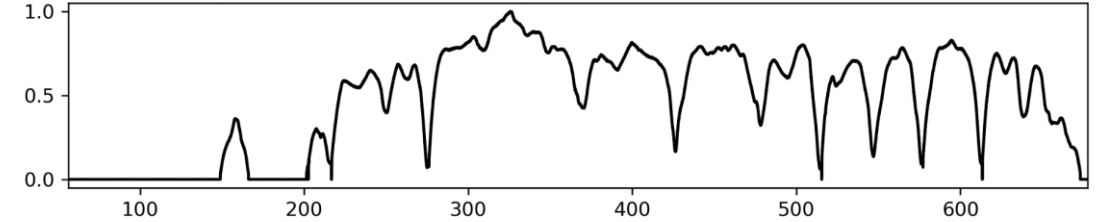
Time Series from Bit Positions 32 to 47 of Arb ID 0x215



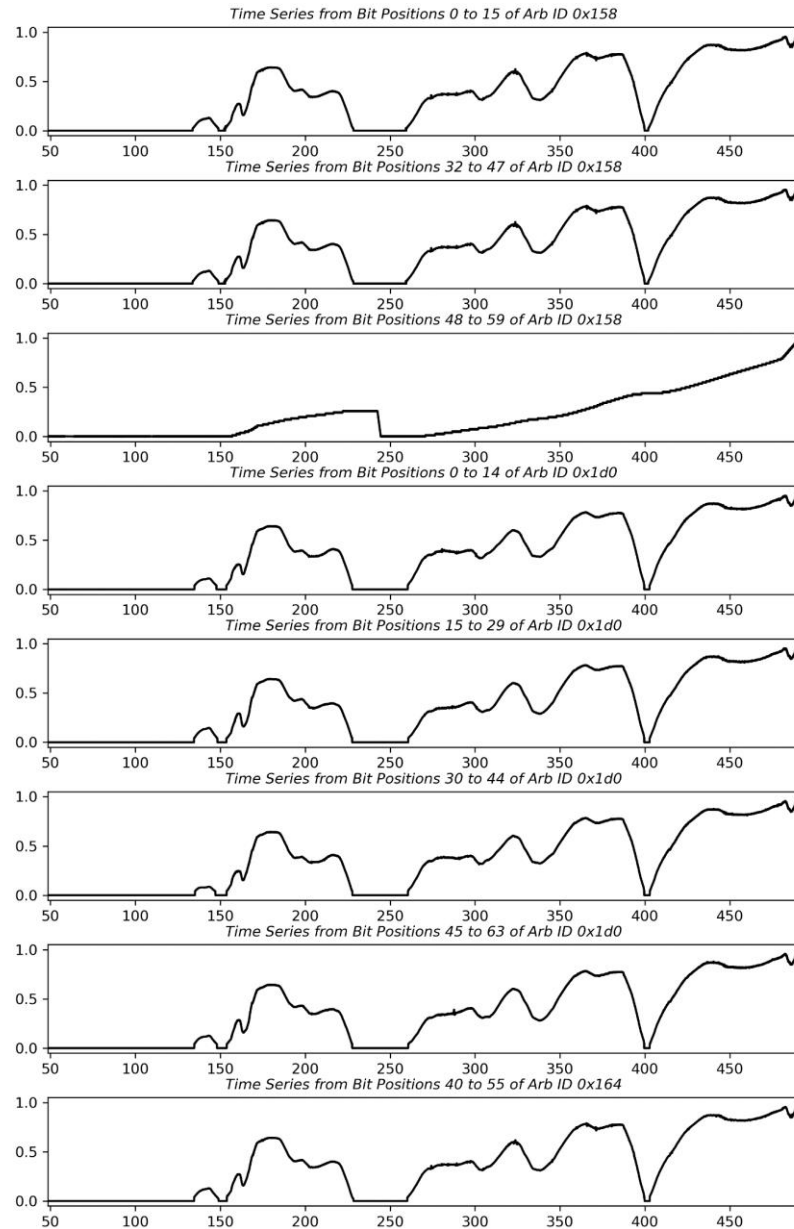
Time Series from Bit Positions 48 to 63 of Arb ID 0x215



Time Series from Bit Positions 16 to 47 of Arb ID 0x201

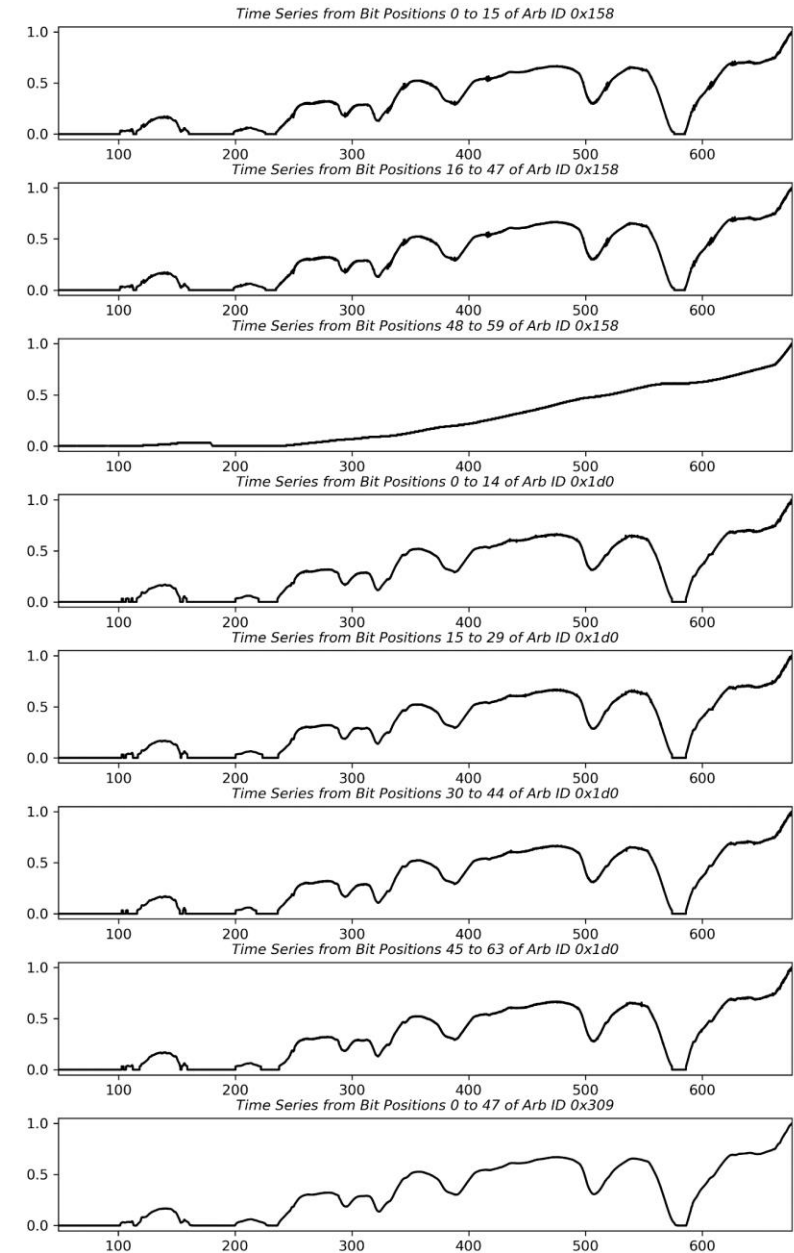


VEHICLE 7



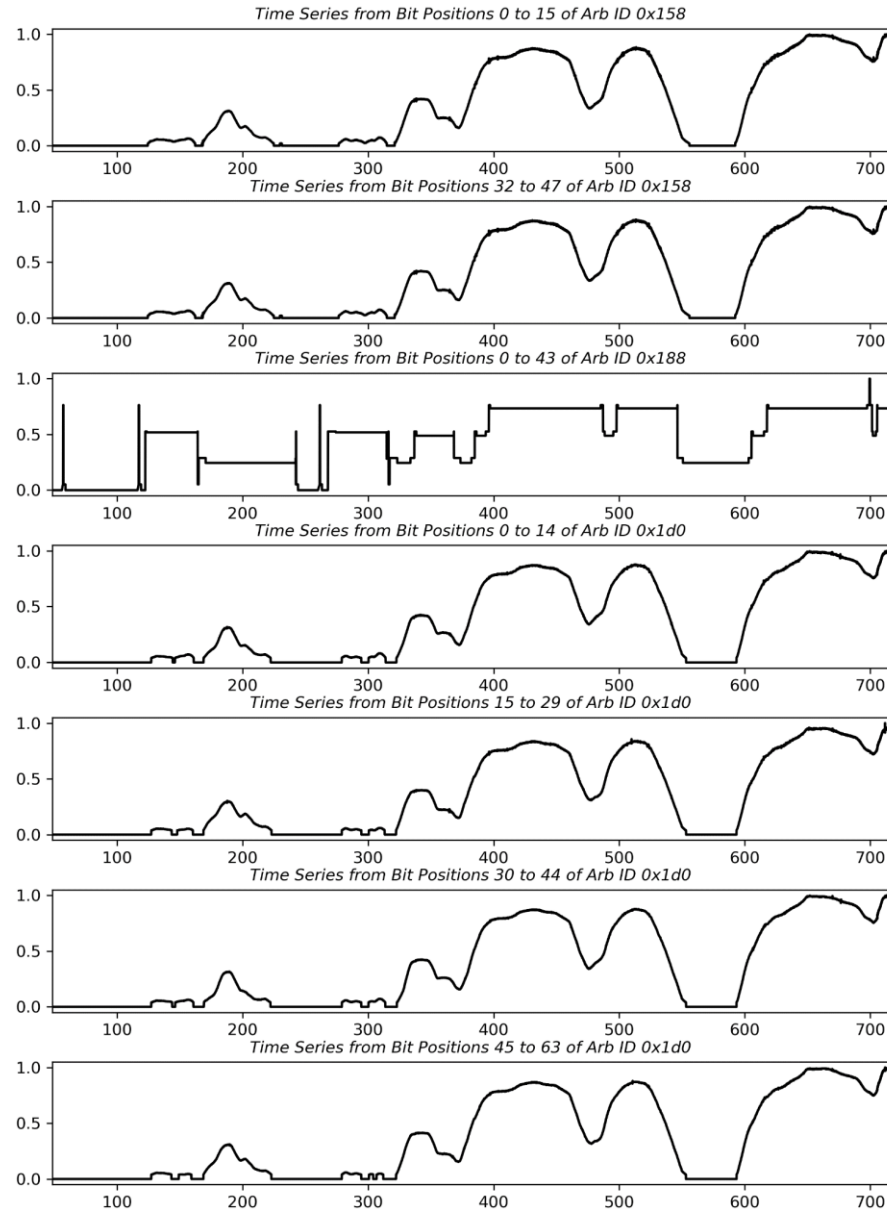
**CROPPED
TO FIT ON
SLIDE**

VEHICLE 8



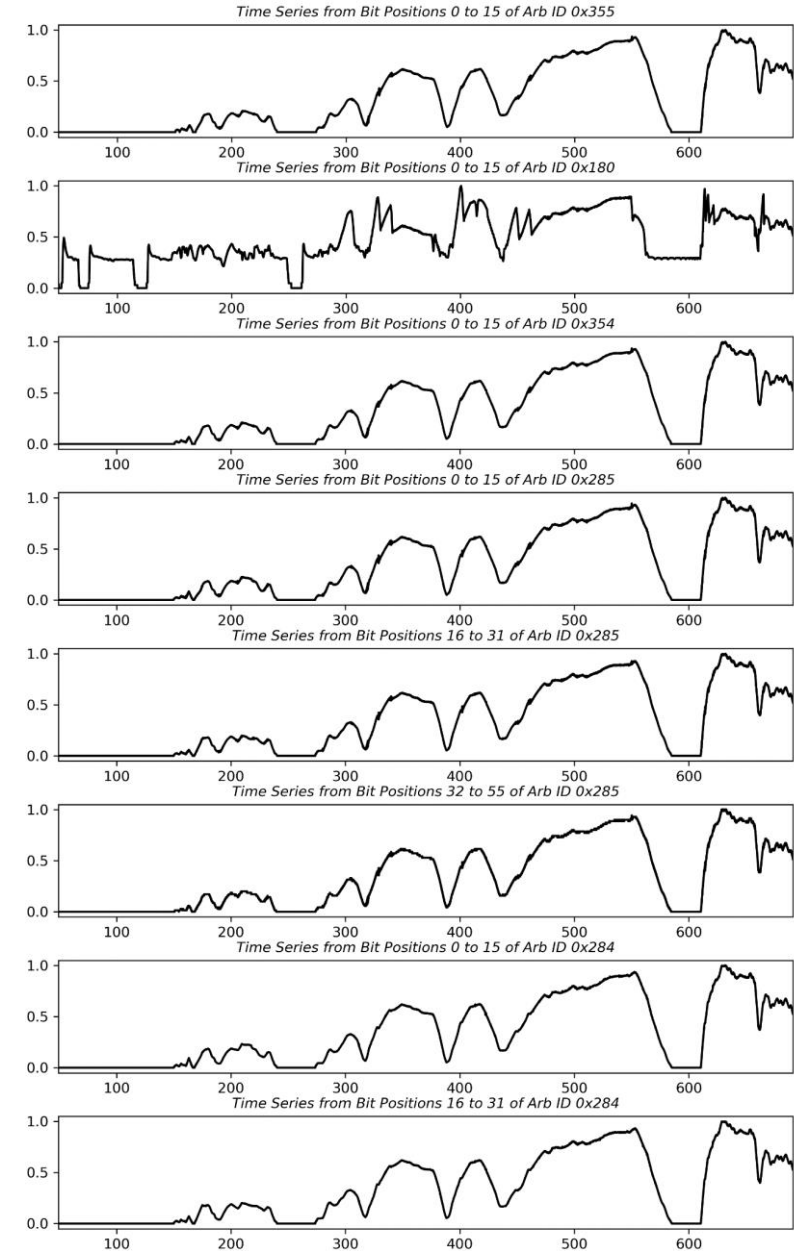
VEHICLE 9

Signal Cluster 17 from Vehicle 8



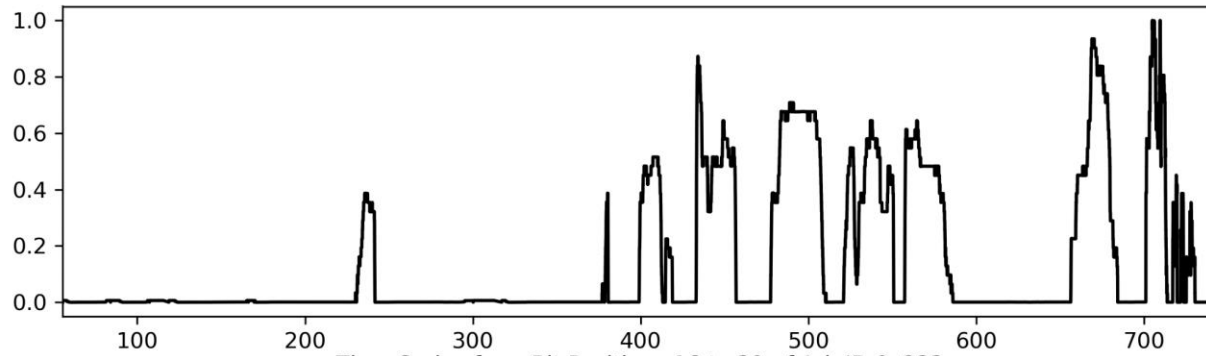
**CROPPED
TO FIT ON
SLIDE**

VEHICLE 10

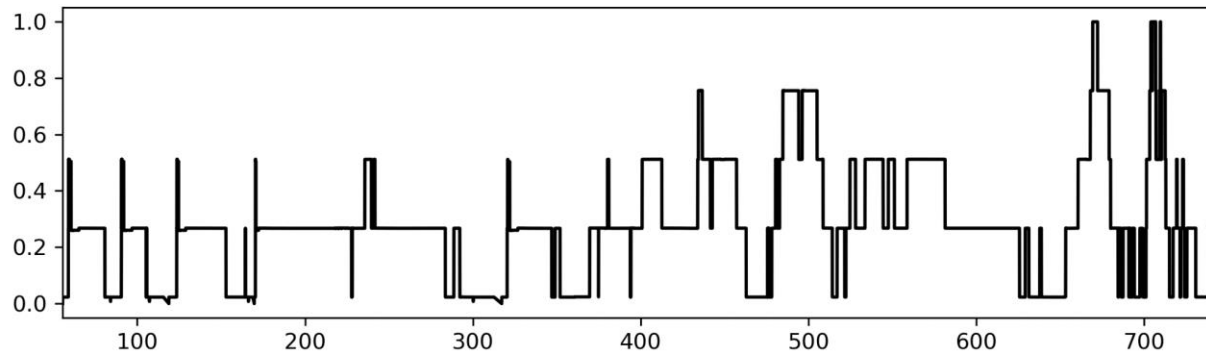


VEHICLE 11

Time Series from Bit Positions 12 to 63 of Arb ID 0x231

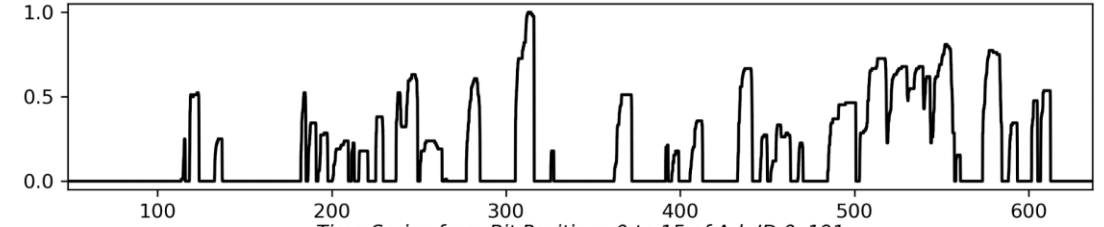


Time Series from Bit Positions 16 to 39 of Arb ID 0x233

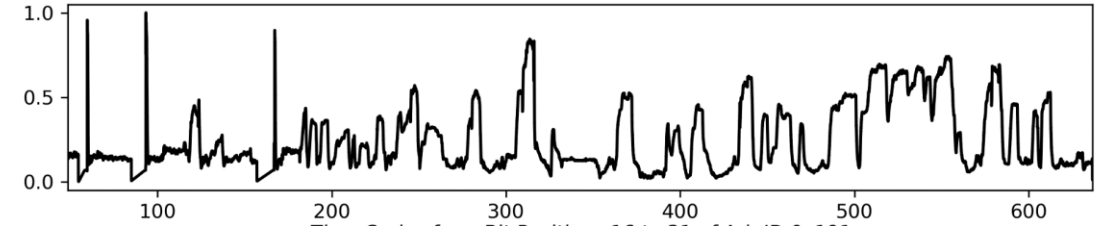


VEHICLE 12

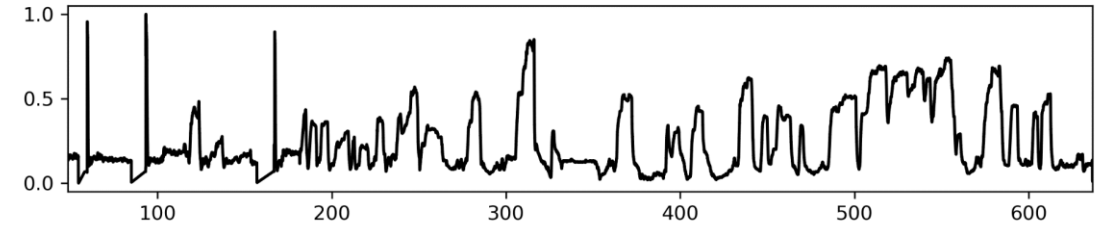
Time Series from Bit Positions 32 to 55 of Arb ID 0xc9



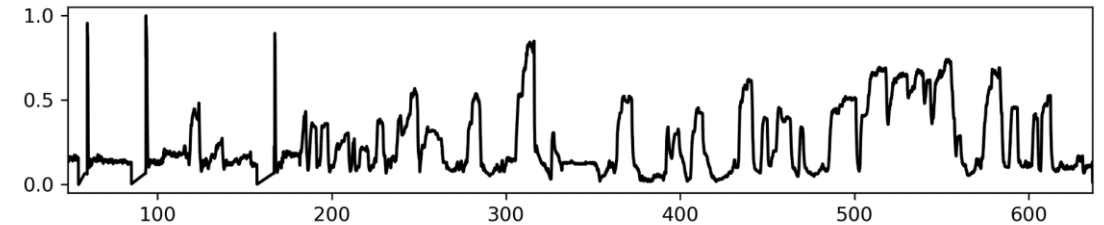
Time Series from Bit Positions 0 to 15 of Arb ID 0x191



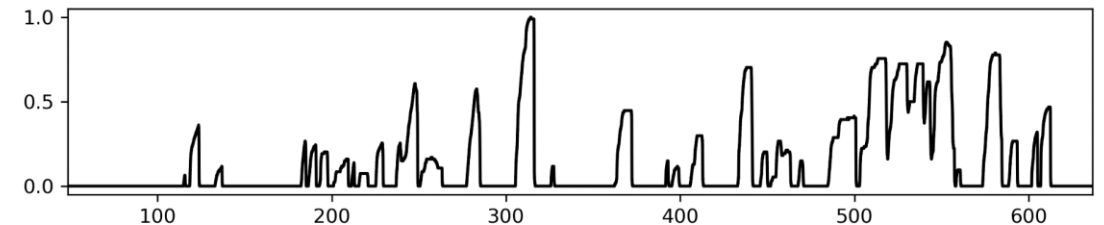
Time Series from Bit Positions 16 to 31 of Arb ID 0x191



Time Series from Bit Positions 32 to 47 of Arb ID 0x191

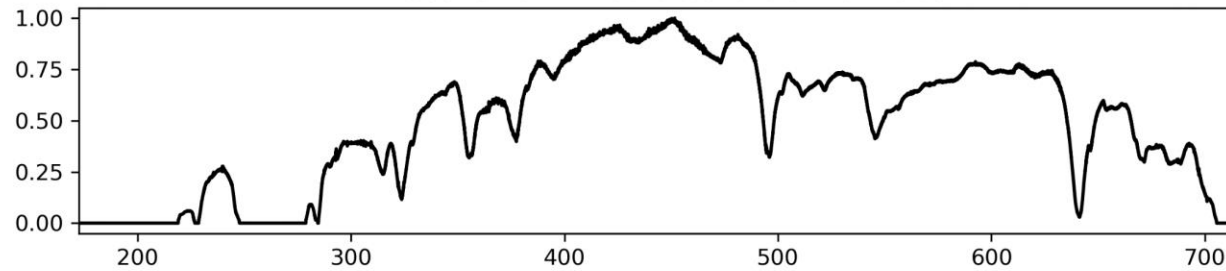


Time Series from Bit Positions 48 to 63 of Arb ID 0x191

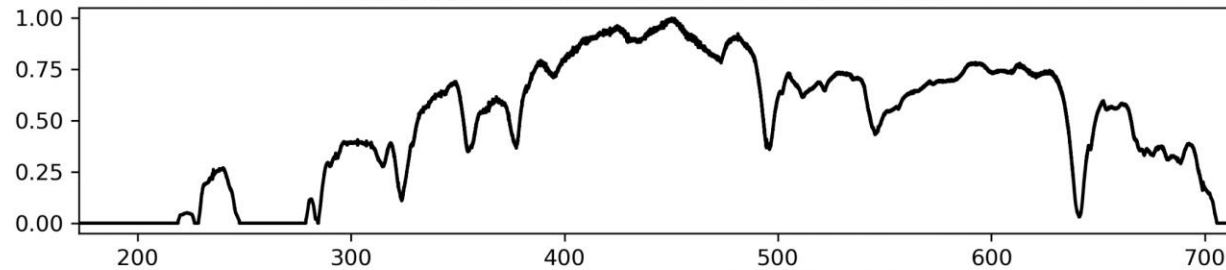


VEHICLE 13

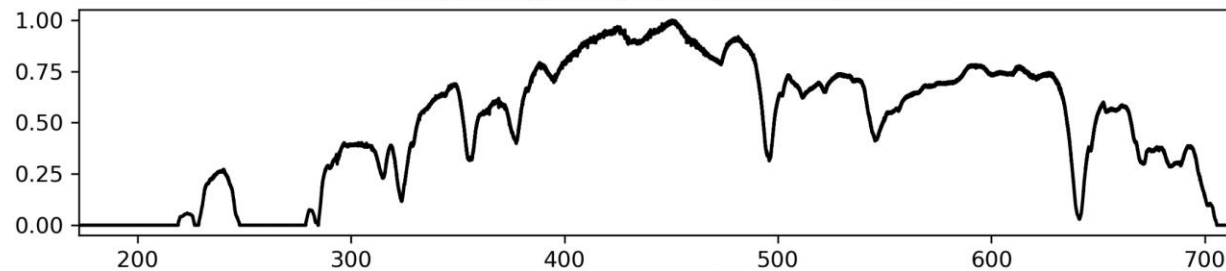
Time Series from Bit Positions 0 to 15 of Arb ID 0xb0



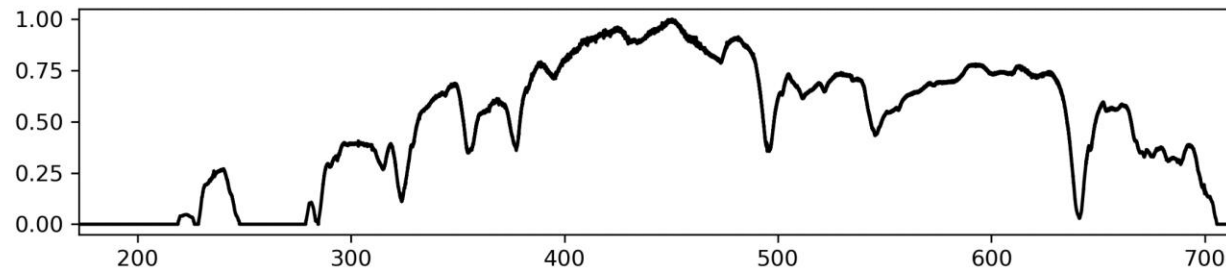
Time Series from Bit Positions 16 to 31 of Arb ID 0xb0



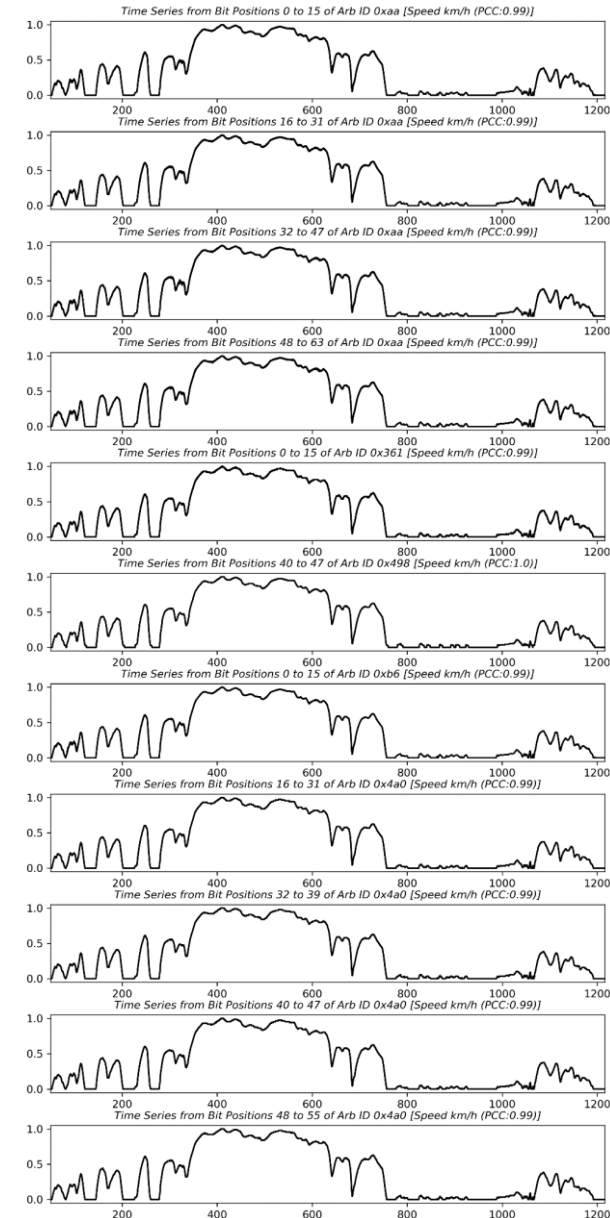
Time Series from Bit Positions 0 to 15 of Arb ID 0xb2



Time Series from Bit Positions 16 to 31 of Arb ID 0xb2

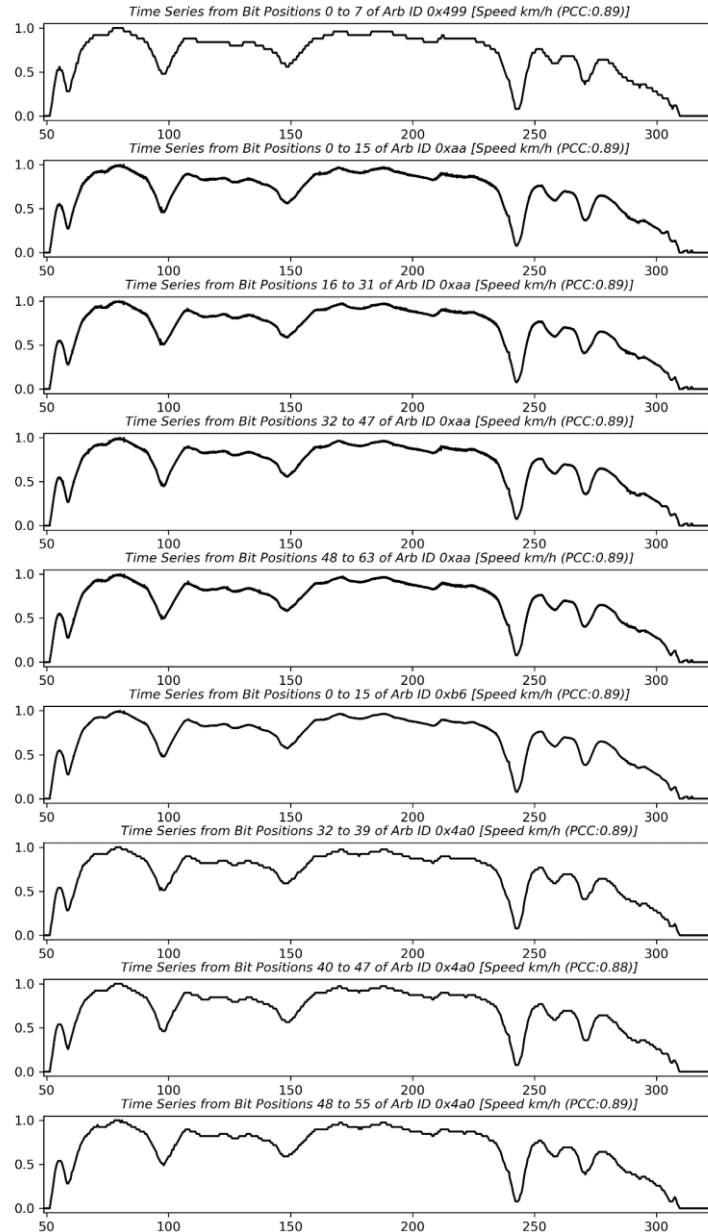


VEHICLE 14

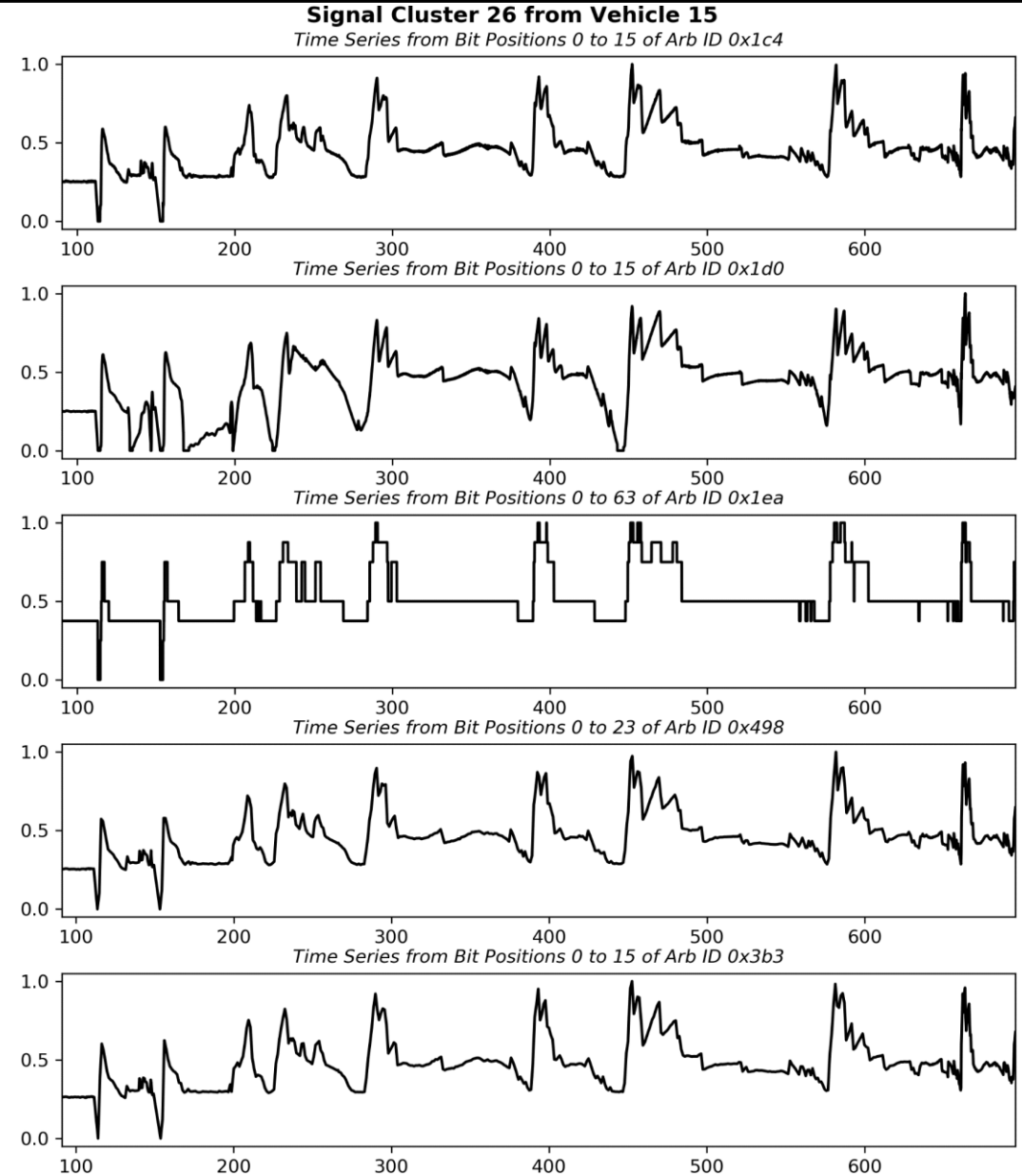


**CROPPED
TO FIT ON
SLIDE**

VEHICLE 15



VEHICLE 16

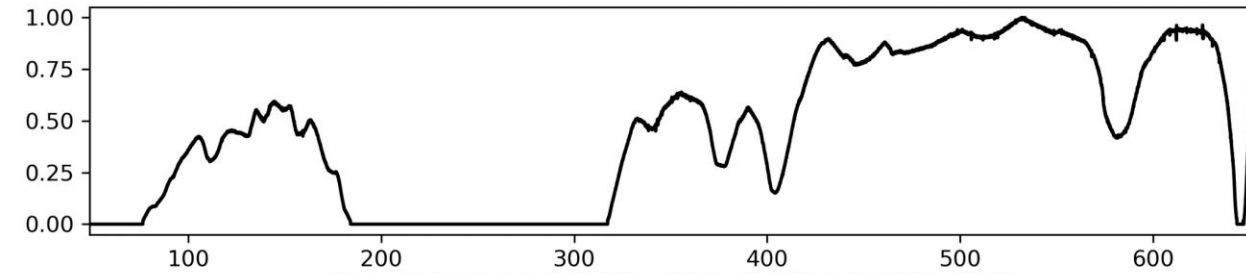


**CROPPED
TO FIT ON
SLIDE**

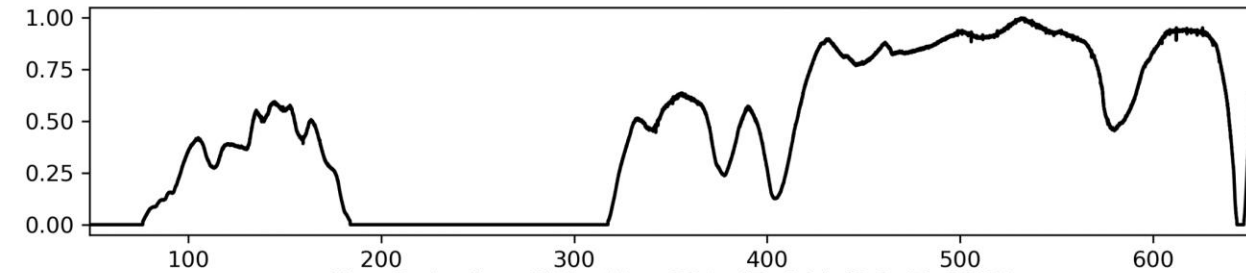
https://github.com/brent-stone/CAN_Reverse_Engineering

VEHICLE 17

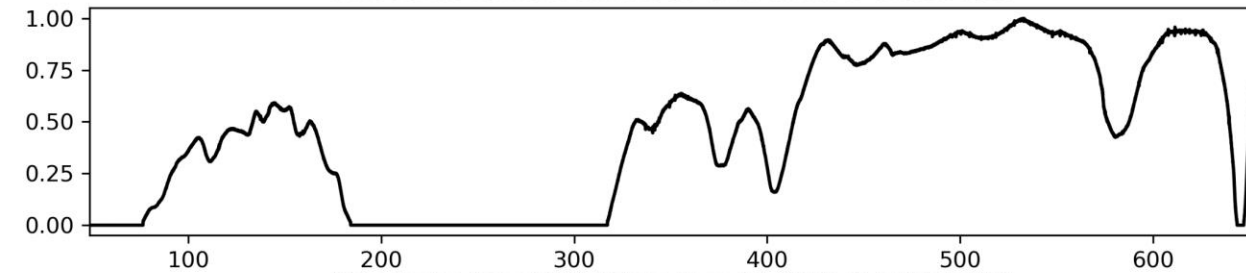
Time Series from Bit Positions 0 to 15 of Arb ID 0x81c04026



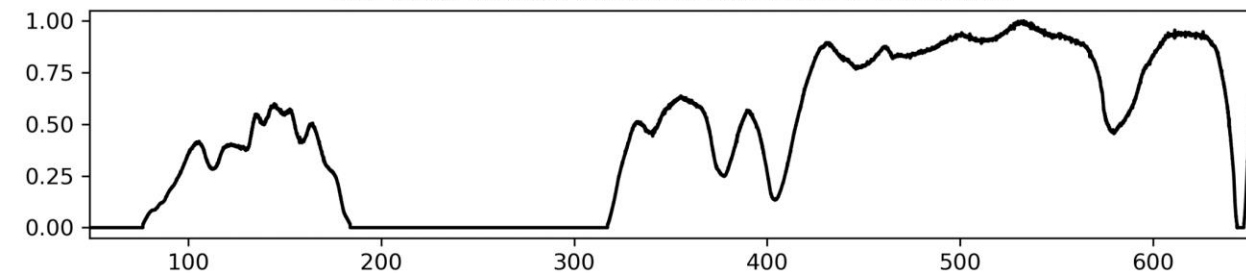
Time Series from Bit Positions 16 to 31 of Arb ID 0x81c04026



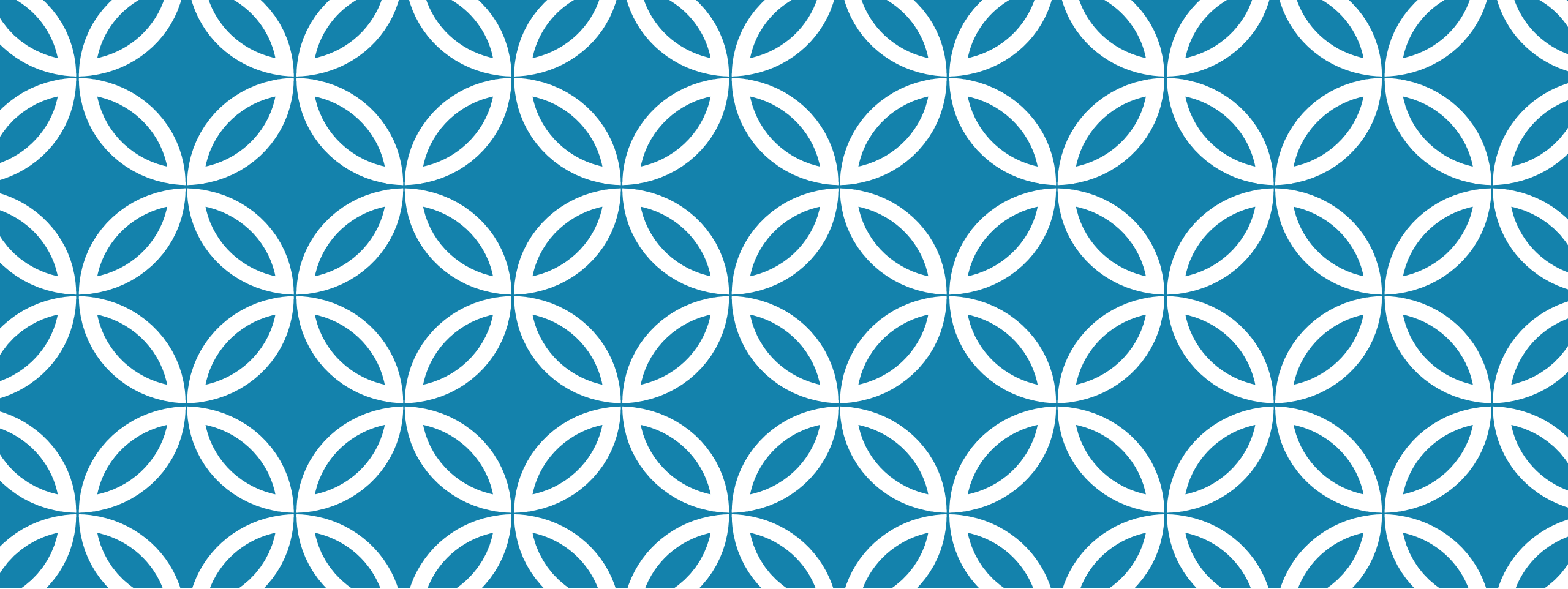
Time Series from Bit Positions 32 to 47 of Arb ID 0x81c04026



Time Series from Bit Positions 48 to 63 of Arb ID 0x81c04026



https://github.com/brent-stone/CAN_Reverse_Engineering



QUESTIONS



**BRENT
STONE**